
MGSC 706: Management Research Statistics

Lecture Notes

PhD Core Course in Management Research Statistics

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These notes cover the course lecture material for the PhD core course in management research statistics.

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Contents

1	Lecture 3: Maximum Likelihood Estimation	2
1.1	Executive overview	2
1.1.1	Learning outcomes	2
1.2	Parametric inference and the likelihood principle	2
1.2.1	From an observed sample to an unknown parameter	2
1.2.2	The three-step maximum likelihood method	3
1.3	Bernoulli coin-flip model	3
1.3.1	Model, coding, and independence	3
1.3.2	Reduction to the number of heads	4
1.3.3	Analytical maximization	4
1.4	Computation: Excel intuition and Python implementation	5
1.4.1	Excel grid search	5
1.4.2	A compact Bernoulli implementation	6
1.5	Normal likelihood for salary and fuel efficiency	7
1.5.1	Normal model	7
1.5.2	Lecture datasets	7
1.5.3	Safer numerical parameterization	8
1.6	Bias–Variance and Irreducible Error Decomposition for Maximum Likelihood Estimation	9
1.6.1	Motivation	9
1.6.2	The fundamental conditional-expectation decomposition	10
1.6.3	Orthogonality of the unpredictable component	10
1.6.4	Connection with the law of total variance	10
1.6.5	Why is this error irreducible?	10
1.6.6	Introducing a parametric likelihood model	11
1.6.7	Maximum likelihood estimation	11
1.6.8	From parameter estimation to prediction	11
1.6.9	Orthogonality eliminates the cross term	11
1.6.10	Bias–variance decomposition of the model component	12
1.6.11	An alternative derivation using the law of total variance	12
1.6.12	Where Fisher information enters	13
1.6.13	Fisher information and prediction variance	13
1.6.14	Interpretation of Fisher information	13
1.6.15	Example: Normal model	14
1.6.16	Example: Poisson MLE	14
1.6.17	Why MLE nonlinearity does not cause a problem	14
1.6.18	Parameter estimation versus prediction	15
1.6.19	The complete conceptual chain	15
1.7	Estimating Irreducible Error, Model Variance, and Squared Bias in Parametric MLE Models	16
1.7.1	General setup	16
1.7.2	Estimate the parametric model by maximum likelihood	16
1.7.3	Prediction-error decomposition	16
1.7.4	Step 1: Estimate irreducible randomness	17
1.7.5	Step 2: Estimate model variance using Fisher information	17
1.7.6	Step 3: Estimate squared bias	17
1.7.7	Parametric bootstrap estimation of model variance and bias	17

1.7.8	General algorithm	18
1.7.9	Example 1: Normal distribution with known variance.	19
1.7.10	Example 2: Normal distribution with unknown mean and variance	19
1.7.11	Example 3: Poisson distribution	20
1.7.12	Example 4: Bernoulli distribution	20
1.7.13	Example 5: Exponential distribution	20
1.7.14	A model in which nonlinear transformation can create prediction bias	21
1.7.15	Two practical approaches.	27
1.7.16	Approach A: Analytic asymptotic decomposition.	27
1.7.17	Approach B: Parametric-bootstrap decomposition.	27
1.7.18	Final summary	27
1.8	Where does higher-order MLE bias come from?	28
1.8.1	Understanding approximation to score equation	28
1.8.2	Geometric interpretation of the score equation and its root	31
1.8.3	What do the score and Hessian terms mean in a single sample?	34
1.8.4	Why the normal-mean score is linear but the normal-variance score is not	36
1.9	Estimation error, bias, and mean squared error.	39
1.9.1	Estimators vary across samples.	39
1.9.2	Consistency and the root-n rate	39
1.10	Fisher information, curvature, and standard errors	39
1.10.1	Why a sharper likelihood means greater precision	39
1.10.2	Bernoulli standard error and confidence interval	40
1.10.3	Normal-model information	41
1.11	Pooling McGill and Toronto salary samples	41
1.11.1	Common-distribution model	41
1.11.2	Separate means with a common standard deviation.	42
1.12	Reading a maximum-likelihood output table.	43
1.13	Common errors and diagnostic checks	43
1.14	Worked practice problems	44
1.14.1	Another Bernoulli sample.	44
1.14.2	Normal scale denominator	44
1.14.3	Sample-size planning by relative precision	44
1.15	Compact mathematical reference	44
1.16	Case studies: The Maximum Likelihood Principle.	45
1.17	Case Study 1: MLE of a Bernoulli Parameter (Coin Flips)	45
1.17.1	Setting and data.	45
1.17.2	Statistical model.	46
1.17.3	Python implementation	46
1.17.4	Output	47
1.17.5	Interpretation	47
1.18	Case Study 2: MLE of a Normal Distribution (Vehicle Fuel Efficiency).	47
1.18.1	Setting and data.	47
1.18.2	Statistical model.	47
1.18.3	Python implementation	48
1.18.4	Output	48
1.18.5	Interpretation	49
1.19	Case Study 3: Confidence Intervals from the Observed Fisher Information (Graduate Salaries).	49
1.19.1	Setting and data.	49
1.19.2	Statistical model.	49
1.19.3	Python implementation	49
1.19.4	Output	51
1.19.5	Interpretation	51
1.20	Case Study 4: A Pooled (Joint) Likelihood Across Two Independent Samples	51
1.20.1	Setting and data.	51

1.20.2	Statistical model	52
1.20.3	Python implementation	52
1.20.4	Output	53
1.20.5	Interpretation	53
1.21	Summary Table	53
1.22	Discussion Questions for Students	53
1.23	Closing synthesis	54

MGSC 706 is a core doctoral course in management research statistics. Its scope proceeds from probability to sampling, statistical inference, hypothesis testing, regression, and model building. Although formulas are essential, the course emphasizes business applications and interpretation. Our plan is to demonstrate Python in a cloud environment, while allowing students to use other statistical platforms such as R or SPSS when appropriate.

The pedagogical principle is learning by doing. Manual calculations are useful for understanding definitions, but real research datasets require software. The deeper goal is to learn how a statistical quantity maps to a research question, how to select the correct comparison group, and how to interpret output rather than merely generate it.

Assessment note. There will be four assignments that are worth 10% each, the midterm exam is worth 20%, the final exam is worth 30%, and participation is worth 10%.

1 | Lecture 3: Maximum Likelihood Estimation

Likelihood • Log-likelihood • Fisher information • Bias • Consistency • Standard error

1.1 Executive overview

Central idea. Maximum likelihood estimation fixes the observed data and chooses the parameter value under which those data would be most plausible.

The lecture opens Chapter 3 by moving from descriptive calculations to parametric inference. It develops the maximum likelihood principle through a coin-flip model, implements the method in Excel and Python, extends it to normal models for MBA salary and automobile fuel-efficiency data, and closes with estimation error, bias, consistency, Fisher information, and standard errors.

1.1.1 Learning outcomes

1. Distinguish probability from likelihood by identifying what is variable and what is held fixed.
2. Construct and maximize Bernoulli and normal likelihood functions.
3. Explain why log-likelihoods are computationally preferable to products of probabilities or densities.
4. Interpret coefficient estimates, standard errors, confidence intervals, and convergence output.
5. State bias, consistency, Fisher information, and the root-n rate precisely.
6. Recognize when pooling groups improves precision and when it imposes an unjustified modeling assumption.

1.2 Parametric inference and the likelihood principle

1.2.1 From an observed sample to an unknown parameter

Let $X_1, \dots, X_n$ denote independent observations generated from a model indexed by an unknown parameter θ . Parametric inference assumes that the data distribution belongs

to a specified family $f(x; \theta)$. The objective is not to alter the realized data; it is to learn which value of θ makes those realized observations most compatible with the model.

$$X_i \stackrel{iid}{\sim} f(x; \theta), \quad i = 1, \dots, n$$

Equation 1. Generic parametric model

Table 1. Probability and likelihood use the same model in different directions

Object	Held fixed	Allowed to vary	Question answered
Probability / density $f(x; \theta)$	Parameter θ	Possible data x	If θ were known, how would outcomes vary?
Likelihood $L(\theta; x)$	Observed data x	Candidate parameter θ	Which θ makes the observed data most plausible?

Interpretive caution. The likelihood is not a probability distribution over the parameter unless a prior distribution and Bayes' theorem are introduced. It ranks parameter values relative to the fixed data.

1.2.2 The three-step maximum likelihood method

1. Specify a parametric model $f(x; \theta)$ and identify the admissible parameter space Θ .
2. Evaluate the joint likelihood of the observed sample as a function of θ .
3. Choose any parameter value that maximizes the likelihood, or equivalently its logarithm.

$$\hat{\theta} \in \arg \max_{\theta \in \Theta} L(\theta; x)$$

Equation 2. Maximum likelihood estimator

1.3 Bernoulli coin-flip model

1.3.1 Model, coding, and independence

The lecture codes a head as $X_i = 1$ and a tail as $X_i = 0$. With $p = P(X_i = 1)$, each trial follows a Bernoulli distribution. Independence means that the outcome of one flip does not change the probability law of another; it does not imply that the coin is fair.

$$X_i \in \{0, 1\}, \quad P(X_i = 1) = p, \quad P(X_i = 0) = 1 - p$$

Equation 3. Bernoulli parameterization

For one observation x_i , both cases can be written compactly as $p^{x_i}(1-p)^{1-x_i}$. Independence turns the joint probability of the ordered sequence into a product.

$$L(p; x) = \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i}$$

Equation 4. Bernoulli likelihood for an ordered sample

1.3.2 Reduction to the number of heads

Let S be the number of heads. The lecture sample has $n = 10$ and $S = 6$. Because exponents add when identical factors are multiplied, every ordered sequence with six heads and four tails has the same parameter-dependent likelihood.

$$S = \sum_{i=1}^n x_i, \quad L(p) = p^S (1-p)^{n-S} = p^6 (1-p)^4$$

Equation 5. Likelihood for six heads and four tails

Table 2. Candidate values from the lecture, recomputed with the corrected exponents

Candidate p	Likelihood $p^6(1-p)^4$	Log-likelihood	Relative conclusion
0.11	0.000001112	-13.7098	Very weak fit
0.40	0.000530842	-7.5410	Below the maximum
0.50	0.000976563	-6.9315	Closer
0.60	0.001194394	-6.7301	Largest of these values

The live spreadsheet initially reversed the exponents before correcting them. The table consistently uses six heads and four tails.

Sequence versus count. If the observed data are only the count $S = 6$ rather than the exact order, the binomial likelihood includes the factor $C(10,6)$. That factor does not depend on p , so it changes the likelihood's height but not its maximizing value.

1.3.3 Analytical maximization

A grid search in Excel illustrates the idea, but calculus provides the exact answer. Because the natural logarithm is strictly increasing, maximizing $L(p)$ is equivalent to maximizing the log-likelihood. The logarithm also converts products into sums and avoids numerical underflow in large samples.

$$\mathcal{L}(p) = \ln L(p) = S \ln p + (n - S) \ln(1 - p)$$

Equation 6. Bernoulli log-likelihood

$$\frac{d\mathcal{L}}{dp} = \frac{S}{p} - \frac{n-S}{1-p} = 0$$

Equation 7. First-order condition

$$\hat{p} = \frac{S}{n} = \frac{6}{10} = 0.60$$

Equation 8. Closed-form Bernoulli MLE

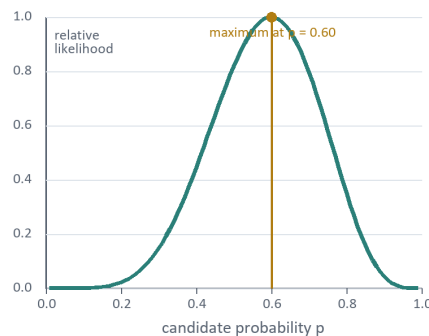
$$\frac{d^2\mathcal{L}}{dp^2} = -\frac{S}{p^2} - \frac{n-S}{(1-p)^2} < 0$$

Equation 9. Concavity verifies a maximum

Boundary cases require separate handling: if $S = 0$, the MLE is $\hat{p} = 0$; if $S = n$, it is $\hat{p} = 1$. Artificially excluding the boundaries with a small epsilon is a numerical convenience, not a change to the mathematical parameter space $[0,1]$.

Likelihood identifies the estimate; information determines its precision

A. Likelihood for $S = 6, n = 10$



B. Approximate 95% intervals

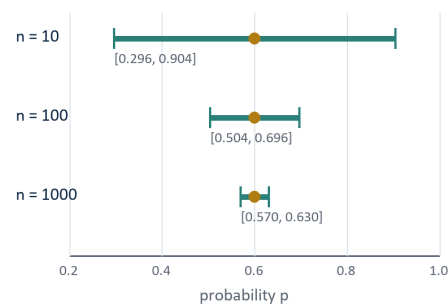


Figure 1. The likelihood locates the point estimate; curvature and sample size govern uncertainty.

Calculated from the lecture's six-head, four-tail Bernoulli example. Intervals use the plug-in Wald approximation.

1.4 Computation: Excel intuition and Python implementation

1.4.1 Excel grid search

The lecture uses an Excel Data Table to evaluate $L(p)$ over a grid from 0 to 1 and then plots the resulting likelihood curve. This is an effective visualization, but the reported maximizer is only as precise as the grid. With increments of 0.10, the search cannot

distinguish 0.60 from a maximizer such as 0.57. A finer grid improves approximation but does not replace analytical or numerical optimization.

1.4.2 A compact Bernoulli implementation

For this model the closed-form estimator is preferable, but the accompanying notebook fits it numerically as a check. The class stores the observed flips in a custom attribute `self.y` rather than passing them through `endog` — `GenericLikelihoodModel` is initialized with a one-element placeholder array instead, and `loglike` reads the real data from `self.y`. A small `_eps` margin keeps the search away from the non-differentiable boundary at $p = 0$ or $p = 1$.

```
import pandas as pd
import numpy as np
from statsmodels.base.model import GenericLikelihoodModel
from scipy.stats import bernoulli

df = pd.read_excel("M3 Coin flip data.xlsx", sheet_name="Sheet1")
data = df["Coins"].to_numpy()          # 100 coded flips (1 = heads, 0 = tails)

class MLE(GenericLikelihoodModel):
    def __init__(self, data):
        self.y = data
        super().__init__(endog=np.zeros(1))
        self.nparams = 1
        self._set_extra_params_names(["p"])
        self._eps = 1e-9
    def loglike(self, params):
        p = params
        if not (self._eps < p < 1 - self._eps):
            return -np.inf
        return bernoulli.logpmf(self.y, p).sum()

model = MLE(data)
result = model.fit(start_params=[0.5], method="bfgs", maxiter=200)
print(result.summary())
```

The optimizer converges in four iterations:

coef	std err	z	P> z	[0.025	0.975]
p	0.6000	0.049	12.247	0.000	0.504 0.696

Log-Likelihood: -67.301 AIC: 136.6 BIC: 134.6

This 100-flip sample happens to share the same sample proportion, $\hat{p} = 0.60$, as the six-head, four-tail toy example used earlier, but at a larger sample size. The reported std err of 0.049 matches Equation 20's plug-in formula exactly,

$$SE(\hat{p}) \approx \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = \sqrt{\frac{0.6 \times 0.4}{100}} \approx 0.049,$$

and lines up with the $n = 100$ row of Table 4.

1.5 Normal likelihood for salary and fuel efficiency

1.5.1 Normal model

The lecture next models continuous observations such as MBA salaries and automobile miles per gallon using a normal distribution. The location parameter μ controls the center, while the positive scale parameter σ controls dispersion.

$$f(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp -\frac{(x - \mu)^2}{2\sigma^2}, \quad \sigma > 0$$

Equation 10. Normal density

$$\mathcal{L}(\mu, \sigma) = -\frac{n}{2} \ln(2\pi) - n \ln \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

Equation 11. Normal log-likelihood

$$\hat{\mu} = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

Equation 12. Closed-form normal MLEs

Important denominator distinction. The normal MLE of variance divides by n . The familiar unbiased sample variance divides by $n - 1$. Therefore `pandas.describe()`, which reports the sample standard deviation by default, need not exactly match the MLE scale estimate.

1.5.2 Lecture datasets

Table 3. Continuous-data illustrations

Dataset	Sample described	Normal-model target	Transcript result / interpretation
MBA salaries	100 observations for each of McGill, Toronto, and UBC	Group mean and standard deviation	For McGill, mean about \$109,028 and scale about \$50,958, confirmed below.
Automobile MPG	Original automobile dataset	Mean MPG and standard deviation	Mean estimate 23.4459 MPG; standard error 0.394, confirmed below.

Fitting the class defined above to each dataset in turn reproduces these numbers exactly:

```
df = pd.read_excel("M3 Salary Data.xlsx", sheet_name="DataTable - 1")
mcgill = df["McGill"].to_numpy()
result_mcgill = MLE(mcgill).fit(start_params=[0, 10], method="bfgs", maxiter=200)
print(result_mcgill.summary())
```

coef	std err	z	P> z	[0.025	0.975]	
mean	1.09e+05	5095.818	21.396	0.000	9.9e+04	1.19e+05
std	5.096e+04	3603.958	14.140	0.000	4.39e+04	5.8e+04

Log-Likelihood: -1225.8 AIC: 2456. BIC: 2452.

so $\hat{\mu} \approx \$109,028$ and $\hat{\sigma} \approx \$50,958$. The identical class applied to the automobile dataset,

```
df = pd.read_excel("M3 Auto_data.xlsx", sheet_name="Original Dataset")
mpg = df["mpg"].to_numpy()
result_mpg = MLE(mpg).fit(start_params=[0, 10], method="bfgs", maxiter=200)
print(result_mpg.summary())
```

coef	std err	z	P> z	[0.025	0.975]	
mean	23.4459	0.394	59.551	0.000	22.674	24.218
std	7.7950	0.278	28.000	0.000	7.249	8.341

Log-Likelihood: -1361.2 AIC: 2726. BIC: 2722.

confirms the mean of 23.4459 MPG and standard error of 0.394 already quoted in Table 3, with $\hat{\sigma} \approx 7.795$ MPG for the scale.

A normal likelihood can be useful even when the empirical salary distribution is visibly skewed, but the assumption should be diagnosed rather than accepted automatically. Under misspecification, the fitted parameters summarize the closest normal approximation in the likelihood sense; standard errors based on the normal model may then need robust adjustment.

1.5.3 Safer numerical parameterization

A direct optimizer can propose a negative σ unless it is constrained. A stable alternative estimates $\eta = \log(\sigma)$, which is unrestricted, and recovers $\sigma = \exp(\eta)$. This avoids discontinuous penalties at zero.

$$\eta = \ln \sigma \in \mathbb{R}, \quad \sigma = e^\eta > 0$$

Equation 13. Log-scale parameterization

In practice, the accompanying notebooks take a simpler route: rather than reparameterizing, `loglike` checks the constraint directly and returns $-\infty$ whenever the proposed σ falls at or below a small threshold `_eps`. This is easier to code and works well once the optimizer is initialized away from the boundary, though the log-scale reparameterization above remains the more robust choice when starting values are less reliable.

```
import numpy as np
```

```

import pandas as pd
from statsmodels.base.model import GenericLikelihoodModel
from scipy.stats import norm

class MLE(GenericLikelihoodModel):
    def __init__(self, data):
        self.y = data
        super().__init__(endog=np.zeros(1))
        self.nparams = 2
        self._set_extra_params_names(["mean", "std"])
        self._eps = 1e-9
    def loglike(self, params):
        mean, std = params
        if not (self._eps < std):
            return -np.inf
        return norm.logpdf(self.y, loc=mean, scale=std).sum()

```

The same class, unchanged, fits every single-sample normal model in this chapter — only the data passed to it and the starting values change, as the next subsection shows.

1.6 Bias–Variance and Irreducible Error Decomposition for Maximum Likelihood Estimation

1.6.1 Motivation

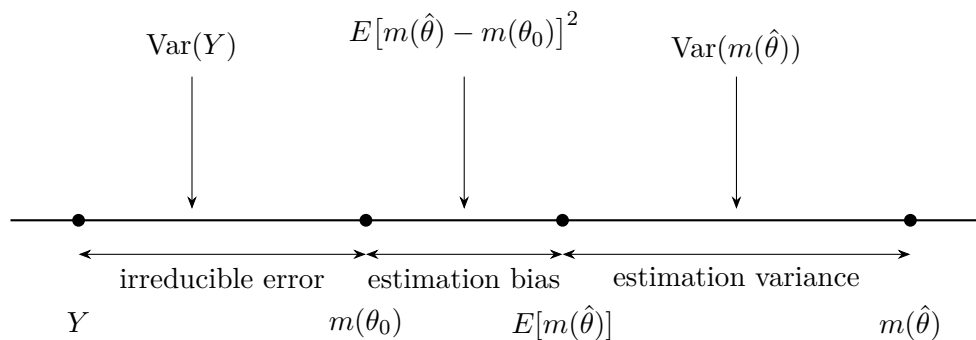


Figure 1.1: Bias–variance decomposition on the number line: the estimand Y , the true conditional mean $m(\theta_0)$, the average fitted value $E[m(\hat{\theta})]$, and a fitted value $m(\hat{\theta})$.

The familiar bias–variance decomposition is often introduced through a regression model $Y = f(X) + \varepsilon$, with $E[\varepsilon | X] = 0$. For a predictor $\hat{f}(X)$, expected squared prediction error is then decomposed into

$$\text{Prediction Error} = \text{Irreducible Error} + \text{Model Variance} + \text{Squared Bias}.$$

At first sight, it is not obvious how this framework applies to maximum likelihood estimation. Maximum likelihood starts from a probability distribution, $Y | X \sim p(y | X; \theta)$, and estimates the unknown parameter θ by maximizing likelihood. The MLE can be a highly nonlinear function of the observations. The key observation is that the

bias–variance decomposition does not fundamentally depend on assuming an additive regression model. The additive representation can be generated from the conditional expectation itself. This allows us to connect three fundamental concepts:

Conditional expectation $\rightarrow$ Irreducible error
 $\rightarrow$ MLE estimation variance through Fisher information.

1.6.2 The fundamental conditional-expectation decomposition

Consider any pair of random variables X and Y , and suppose the relevant moments exist. Define $m(X) = E[Y | X]$. Now define $\varepsilon = Y - m(X)$. Then, simply by rearranging, $Y = m(X) + \varepsilon$. This looks like the standard additive regression model, but there is an important difference. We did not assume that the data-generating mechanism was additive. We constructed this representation by defining $m(X)$ as the true conditional expectation. Consequently, $E[\varepsilon | X] = E[Y - m(X) | X]$. Because $m(X)$ is a function of X , $E[m(X) | X] = m(X)$. Therefore, $E[\varepsilon | X] = E[Y | X] - m(X) = 0$. Hence $E[\varepsilon | X] = 0$. Conditional mean zero is therefore automatic when the residual is defined relative to the true conditional expectation.

1.6.3 Orthogonality of the unpredictable component

The condition $E[\varepsilon | X] = 0$ has an important implication. For any suitable function $g(X)$, $E[\varepsilon g(X)] = E[E[\varepsilon g(X) | X]]$. Since $g(X)$ is known conditional on X , $E[\varepsilon g(X) | X] = g(X)E[\varepsilon | X]$. Therefore, $E[\varepsilon g(X)] = 0$. Thus the residual is orthogonal to every function of X . In particular, $E[\varepsilon m(X)] = 0$. We can therefore interpret $Y = m(X) + \varepsilon$ as an orthogonal decomposition:

$$Y = \underbrace{E[Y | X]}_{\text{predictable component}} + \underbrace{\varepsilon}_{\text{unpredictable component}}.$$

The conditional expectation contains everything about Y 's mean that can be predicted from X . The residual contains the remaining variation.

1.6.4 Connection with the law of total variance

The orthogonal decomposition also explains the law of total variance. Since $Y = m(X) + \varepsilon$ and $\text{Cov}(m(X), \varepsilon) = 0$, we have $\text{Var}(Y) = \text{Var}(m(X)) + \text{Var}(\varepsilon)$. Now $\text{Var}(\varepsilon) = E[\varepsilon^2]$, because $E[\varepsilon] = 0$. Furthermore, $E[\varepsilon^2 | X] = E[(Y - E[Y | X])^2 | X]$. By definition, $E[\varepsilon^2 | X] = \text{Var}(Y | X)$. Taking expectations, $E[\varepsilon^2] = E[\text{Var}(Y | X)]$. Therefore,

$$\text{Var}(Y) = \text{Var}(E[Y | X]) + E[\text{Var}(Y | X)].$$

The first component is variation explained by X . The second is variation remaining after X is known. This second component is the foundation of irreducible prediction error.

1.6.5 Why is this error irreducible?

Suppose we somehow know the true conditional expectation perfectly. Our optimal squared-error predictor is then $m(X) = E[Y | X]$. Nevertheless, $Y = m(X) + \varepsilon$. Therefore the prediction error of this perfect predictor is $Y - m(X) = \varepsilon$. Its conditional

expected squared error is $E[(Y - m(X))^2 | X] = E[\varepsilon^2 | X] = \text{Var}(Y | X)$. Thus Irreducible error at $X = x = \text{Var}(Y | X = x)$. No improvement in estimation can eliminate this component. It represents randomness in a future realization of Y even when the true conditional distribution is known.

1.6.6 Introducing a parametric likelihood model

Now suppose that the conditional distribution belongs to a parametric family: $Y | X = x \sim p(y | x; \theta_0)$, where θ_0 is the true parameter. The model implies a conditional mean $m_{\theta_0}(x) = E_{\theta_0}[Y | X = x]$. If the parametric model is correctly specified, this is the true conditional expectation: $m_{\theta_0}(X) = E[Y | X]$. Consequently, we can write $Y = m_{\theta_0}(X) + \varepsilon$ with $\varepsilon = Y - m_{\theta_0}(X)$ and $E[\varepsilon | X] = 0$. Furthermore, $\text{Var}(\varepsilon | X) = \text{Var}_{\theta_0}(Y | X)$. Notice that we have not assumed an additive likelihood model. We have simply decomposed the random outcome around the conditional mean implied by the likelihood.

1.6.7 Maximum likelihood estimation

Suppose we observe a training sample $D = \{(X_i, Y_i) : i = 1, \dots, n\}$. The likelihood is $L(\theta; D) = \prod_{i=1}^n p(Y_i | X_i; \theta)$, or equivalently the log-likelihood $\ell_n(\theta) = \sum_{i=1}^n \log p(Y_i | X_i; \theta)$. The MLE is $\hat{\theta}_D = \arg \max_{\theta} \ell_n(\theta)$. Because the training dataset D is random, $\hat{\theta}_D$ is random. Different training samples produce different estimates:

$$D_1 \rightarrow \hat{\theta}_1, \quad D_2 \rightarrow \hat{\theta}_2, \quad D_3 \rightarrow \hat{\theta}_3.$$

This sampling variation in the estimated parameter generates model variance.

1.6.8 From parameter estimation to prediction

For a new observation with covariates $X = x$, the true conditional mean is $m_{\theta_0}(x)$. But since we do not know θ_0 , we use $m_{\hat{\theta}_D}(x)$. A new outcome can be written $Y_{\text{new}} = m_{\theta_0}(x) + \varepsilon_{\text{new}}$. Therefore the prediction error is

$$Y_{\text{new}} - m_{\hat{\theta}_D}(x) = \varepsilon_{\text{new}} + m_{\theta_0}(x) - m_{\hat{\theta}_D}(x).$$

This is the crucial decomposition:

$$\text{Prediction error} = \text{Outcome randomness} + \text{Model estimation error.}$$

The first comes from the new observation. The second comes from estimating θ_0 using a finite training sample.

1.6.9 Orthogonality eliminates the cross term

Square the prediction error:

$$\begin{aligned} (Y_{\text{new}} - m_{\hat{\theta}_D}(x))^2 &= \varepsilon_{\text{new}}^2 \\ &\quad + (m_{\theta_0}(x) - m_{\hat{\theta}_D}(x))^2 \\ &\quad + 2\varepsilon_{\text{new}}(m_{\theta_0}(x) - m_{\hat{\theta}_D}(x)). \end{aligned}$$

For an independent new observation, $E[\varepsilon_{\text{new}} \mid X = x, D] = 0$. Therefore the cross term has expectation zero. Hence

$$E[(Y_{\text{new}} - m_{\hat{\theta}_D}(x))^2] = E[\varepsilon_{\text{new}}^2] + E[(m_{\hat{\theta}_D}(x) - m_{\theta_0}(x))^2].$$

The first component is $E[\varepsilon_{\text{new}}^2 \mid X = x] = \text{Var}_{\theta_0}(Y \mid X = x)$. Thus

$$\text{Prediction MSE} = \text{Irreducible Error} + \text{MSE of the estimated conditional mean.}$$

1.6.10 Bias–variance decomposition of the model component

Now apply the ordinary bias–variance identity to $m_{\hat{\theta}_D}(x)$. For any random estimator Z of a fixed target z_0 , $E[(Z - z_0)^2] = \text{Var}(Z) + (E[Z] - z_0)^2$. Therefore,

$$E[(m_{\hat{\theta}_D}(x) - m_{\theta_0}(x))^2] = \text{Var}_D(m_{\hat{\theta}_D}(x)) + \left(E_D[m_{\hat{\theta}_D}(x)] - m_{\theta_0}(x)\right)^2.$$

Combining this with the irreducible-error decomposition gives

$$E[(Y_{\text{new}} - m_{\hat{\theta}_D}(x))^2] = \underbrace{\text{Var}_{\theta_0}(Y \mid X = x)}_{\text{Irreducible Error}} + \underbrace{\text{Var}_D(m_{\hat{\theta}_D}(x))}_{\text{Model Variance}} + \underbrace{\left(E_D[m_{\hat{\theta}_D}(x)] - m_{\theta_0}(x)\right)^2}_{\text{Squared Bias}}.$$

This is the MLE analogue of the familiar regression bias–variance decomposition.

1.6.11 An alternative derivation using the law of total variance

We can also obtain the variance part directly from the law of total variance. Define the prediction error $Z = Y_{\text{new}} - m_{\hat{\theta}_D}(x)$. Condition on the training sample D . Once D is given, $m_{\hat{\theta}_D}(x)$ is fixed. Therefore $\text{Var}(Z \mid D, X = x) = \text{Var}(Y_{\text{new}} \mid X = x)$. Thus $E_D[\text{Var}(Z \mid D, X = x)] = \text{Var}_{\theta_0}(Y \mid X = x)$. This is the irreducible component. Now examine the conditional expectation: $E[Z \mid D, X = x] = m_{\theta_0}(x) - m_{\hat{\theta}_D}(x)$. Therefore, $\text{Var}_D(E[Z \mid D, X = x]) = \text{Var}_D(m_{\hat{\theta}_D}(x))$. Applying the law of total variance,

$$\text{Var}(Z \mid X = x) = E_D[\text{Var}(Z \mid D, X = x)] + \text{Var}_D(E[Z \mid D, X = x]).$$

Hence

$$\text{Var}(Z \mid X = x) = \underbrace{\text{Var}(Y \mid X = x)}_{\text{irreducible variance}} + \underbrace{\text{Var}_D(m_{\hat{\theta}_D}(x))}_{\text{model variance}}.$$

This is precisely the total-variance decomposition of prediction-error variance. The squared bias is then added through $E[Z^2] = \text{Var}(Z) + (E[Z])^2$. Therefore,

$$\text{Prediction MSE} = \text{Irreducible Variance} + \text{Model Variance} + \text{Squared Bias.}$$

1.6.12 Where Fisher information enters

So far nothing has been specifically about maximum likelihood except the construction of $\hat{\theta}_D$. Maximum likelihood theory now gives us a way to characterize the model-variance component. Define the score for one observation: $s(Y, X; \theta) = \frac{\partial}{\partial \theta} \log p(Y | X; \theta)$. The Fisher information matrix for the sample is

$$I_n(\theta_0) = E \left[-\frac{\partial^2 \ell_n(\theta_0)}{\partial \theta \partial \theta'} \right].$$

Under the usual regularity conditions, $\hat{\theta}_{\text{MLE}} \approx N(\theta_0, I_n(\theta_0)^{-1})$. Thus $\text{Var}(\hat{\theta}_{\text{MLE}}) \approx I_n(\theta_0)^{-1}$. For iid observations, $I_n(\theta_0) = nI_1(\theta_0)$, so $\text{Var}(\hat{\theta}_{\text{MLE}}) \approx \frac{1}{n}I_1(\theta_0)^{-1}$. Fisher information therefore characterizes how much the estimated parameter changes across different training samples.

1.6.13 Fisher information and prediction variance

However, prediction is based not directly on $\hat{\theta}$, but on $m_{\hat{\theta}}(x)$. If this function is nonlinear in θ , use a first-order Taylor expansion around θ_0 :

$$m_{\hat{\theta}}(x) \approx m_{\theta_0}(x) + \nabla_{\theta} m_{\theta_0}(x)' (\hat{\theta} - \theta_0).$$

Therefore, by the delta method,

$$\text{Var}(m_{\hat{\theta}}(x)) \approx \nabla_{\theta} m_{\theta_0}(x)' \text{Var}(\hat{\theta}) \nabla_{\theta} m_{\theta_0}(x).$$

Substituting the MLE covariance,

$$\text{Var}(m_{\hat{\theta}}(x)) \approx \nabla_{\theta} m_{\theta_0}(x)' I_n(\theta_0)^{-1} \nabla_{\theta} m_{\theta_0}(x).$$

This is the MLE's model-variance component. Hence the approximate prediction decomposition becomes

$$\begin{aligned} \text{Prediction MSE}(x) \approx & \underbrace{\text{Var}_{\theta_0}(Y | X = x)}_{\text{Irreducible Error}} \\ & + \underbrace{\nabla_{\theta} m_{\theta_0}(x)' I_n(\theta_0)^{-1} \nabla_{\theta} m_{\theta_0}(x)}_{\text{MLE Model Variance}} \\ & + \underbrace{\text{Bias}(m_{\hat{\theta}}(x))^2}_{\text{Squared Bias}}. \end{aligned}$$

This equation connects bias–variance decomposition directly to Fisher information.

1.6.14 Interpretation of Fisher information

The distinction between the first two terms is fundamental. The quantity $\text{Var}_{\theta_0}(Y | X = x)$ describes uncertainty in a new observation even if θ_0 were known perfectly. By contrast,

$I_n(\theta_0)^{-1}$ describes uncertainty because θ_0 is unknown and must be estimated from finite data. Thus:

Source of uncertainty	Quantity	Can more data eliminate it?
Outcome randomness	$\text{Var}(Y \mid X)$	No
Parameter estimation	$I_n(\theta_0)^{-1}$	Yes

As n increases, $I_n(\theta_0) = nI_1(\theta_0)$, so $I_n(\theta_0)^{-1} = \frac{1}{n}I_1(\theta_0)^{-1} \rightarrow 0$. Thus model variance disappears asymptotically. But $\text{Var}(Y \mid X = x)$ does not disappear. This is exactly why it is called irreducible error.

1.6.15 Example: Normal model

Suppose $Y_i \sim N(\mu, \sigma^2)$, with σ^2 known. There are no covariates, so $E[Y] = \mu$. The orthogonal decomposition is simply $Y = \mu + \varepsilon$, where $E[\varepsilon] = 0$, $\text{Var}(\varepsilon) = \sigma^2$. The MLE is $\hat{\mu} = \bar{Y}$. For one observation, Fisher information for μ is $I_1(\mu) = \frac{1}{\sigma^2}$. For n observations, $I_n(\mu) = \frac{n}{\sigma^2}$. Therefore, $I_n(\mu)^{-1} = \frac{\sigma^2}{n}$. Indeed, $\text{Var}(\hat{\mu}) = \frac{\sigma^2}{n}$. Now consider a new observation $Y_{\text{new}} = \mu + \varepsilon_{\text{new}}$. Prediction error is $Y_{\text{new}} - \hat{\mu} = \varepsilon_{\text{new}} + (\mu - \hat{\mu})$. The two terms are independent because the new observation is independent of the training sample. Consequently, $E[(Y_{\text{new}} - \hat{\mu})^2] = \sigma^2 + \frac{\sigma^2}{n}$. Thus

$$\text{Prediction MSE} = \underbrace{\sigma^2}_{\text{irreducible}} + \underbrace{\frac{\sigma^2}{n}}_{\text{MLE variance}} + \underbrace{0}_{\text{squared bias}}.$$

Using Fisher information, $\boxed{\text{Prediction MSE} = \sigma^2 + I_n(\mu)^{-1}}$. This is the simplest possible example of the general result.

1.6.16 Example: Poisson MLE

Consider $Y \sim \text{Poisson}(\lambda)$. Then $E[Y] = \lambda$ and $\text{Var}(Y) = \lambda$. We can therefore write $Y = \lambda + \varepsilon$ where $\varepsilon = Y - \lambda$. Automatically, $E[\varepsilon] = 0$ and $E[\varepsilon^2] = \lambda$. Thus the irreducible error for predicting a new Poisson realization is $\boxed{\lambda}$. For n iid observations, $\hat{\lambda}_{\text{MLE}} = \bar{Y}$. The Fisher information in one observation is $I_1(\lambda) = \frac{1}{\lambda}$. Hence $I_n(\lambda) = \frac{n}{\lambda}$ and $I_n(\lambda)^{-1} = \frac{\lambda}{n}$. Therefore, $\text{Var}(\hat{\lambda}) = \frac{\lambda}{n}$. Since the estimator is unbiased, $E[\hat{\lambda}] = \lambda$. Prediction of a new observation therefore gives

$$E[(Y_{\text{new}} - \hat{\lambda})^2] = \underbrace{\lambda}_{\text{irreducible}} + \underbrace{\frac{\lambda}{n}}_{\text{model variance}} + \underbrace{0}_{\text{bias}^2}.$$

Again, Fisher information determines the estimation component but not the irreducible component.

1.6.17 Why MLE nonlinearity does not cause a problem

MLE is often a nonlinear function of the observations: $\hat{\theta} = h(D)$. Nothing in the bias–variance identity requires the estimator to be linear. For any estimator,

$$E[(\hat{\theta} - \theta_0)^2] = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2.$$

Likewise, for any prediction function $m_{\hat{\theta}}(x)$,

$$E[(m_{\hat{\theta}}(x) - m_{\theta_0}(x))^2] = \text{Var}(m_{\hat{\theta}}(x)) + \text{Bias}(m_{\hat{\theta}}(x))^2.$$

Nonlinearity merely makes these quantities harder to calculate exactly. MLE theory solves this difficulty asymptotically: $\hat{\theta} \approx N(\theta_0, I_n^{-1})$, and the delta method propagates this uncertainty through the nonlinear function $m_{\theta}(x)$. Thus Fisher information and the delta method provide the bridge from nonlinear MLE estimation to the familiar bias–variance framework.

1.6.18 Parameter estimation versus prediction

It is important not to confuse these two problems. If our target is the parameter itself, θ_0 , then

$$E[(\hat{\theta} - \theta_0)^2] = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2.$$

There is no irreducible-error term because θ_0 is a fixed parameter. For a regular MLE, $\text{Var}(\hat{\theta}) \approx I_n(\theta_0)^{-1}$. By contrast, if our target is a new random observation Y_{new} , then there is additional outcome randomness. Therefore,

$$\boxed{\text{Parameter-estimation MSE} = \text{Variance} + \text{Bias}^2} \text{ whereas}$$

$$\boxed{\text{Prediction MSE} = \text{Irreducible Error} + \text{Model Variance} + \text{Bias}^2}.$$

This distinction explains why irreducible error appears in prediction but not in estimation of a fixed parameter.

1.6.19 The complete conceptual chain

The entire argument can now be summarized in one sequence. Start from the true conditional distribution $Y | X \sim p(y | X; \theta_0)$. It implies the conditional mean $m_{\theta_0}(X) = E[Y | X]$. Therefore, $Y = m_{\theta_0}(X) + \varepsilon$, $E[\varepsilon | X] = 0$. The conditional residual variance is $\text{Var}(\varepsilon | X) = \text{Var}(Y | X)$, which gives the irreducible error. Because θ_0 is unknown, MLE estimates it: $\theta_0 \rightarrow \hat{\theta}_D$. This changes the conditional-mean predictor from $m_{\theta_0}(X)$ to $m_{\hat{\theta}_D}(X)$. Different training datasets produce different $\hat{\theta}_D$'s, generating model variance. MLE theory tells us $\text{Var}(\hat{\theta}_D) \approx I_n(\theta_0)^{-1}$. The delta method then gives

$$\text{Var}(m_{\hat{\theta}_D}(x)) \approx \nabla m_{\theta_0}(x)' I_n(\theta_0)^{-1} \nabla m_{\theta_0}(x).$$

Finally, any systematic difference between the average fitted conditional mean and the true conditional mean produces bias. Therefore,

$$\boxed{\begin{aligned} \text{Prediction MSE}(x) \approx & \underbrace{\text{Var}_{\theta_0}(Y | X = x)}_{\text{irreducible randomness}} \\ & + \underbrace{\nabla m_{\theta_0}(x)' I_n(\theta_0)^{-1} \nabla m_{\theta_0}(x)}_{\text{finite-sample estimation variance}} \\ & + \underbrace{(E[m_{\hat{\theta}}(x)] - m_{\theta_0}(x))^2}_{\text{squared bias}}. \end{aligned}}$$

The first term belongs to the **outcome distribution**. The second term belongs to the **estimation procedure** and is governed asymptotically by Fisher information. The third

term measures **systematic estimation error**. As sample size grows, the second term typically shrinks at rate $1/n$, and under regularity the MLE bias also vanishes. The first term remains. That persistent first component is the irreducible error.

1.7 Estimating Irreducible Error, Model Variance, and Squared Bias in Parametric MLE Models

1.7.1 General setup

Suppose $Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} p(y; \theta_0)$, where θ_0 is the unknown true parameter. For the moment, there are no covariates X . Define the population mean implied by the parametric model as $\mu(\theta) = E_\theta[Y]$, and define the population variance as $V(\theta) = \text{Var}_\theta(Y)$. Under the true parameter, $\mu_0 = \mu(\theta_0)$, $V_0 = V(\theta_0)$. A future observation can always be decomposed as $Y_{\text{new}} = \mu_0 + \varepsilon_{\text{new}}$, where $\varepsilon_{\text{new}} = Y_{\text{new}} - \mu_0$. By construction, $E[\varepsilon_{\text{new}}] = 0$ and $E[\varepsilon_{\text{new}}^2] = V_0$. Thus, $V_0 = \text{Var}_{\theta_0}(Y)$ is the irreducible randomness in predicting a new observation.

1.7.2 Estimate the parametric model by maximum likelihood

Given observations $D = (Y_1, \dots, Y_n)$, define the likelihood $L_n(\theta) = \prod_{i=1}^n p(Y_i; \theta)$ and log-likelihood $\ell_n(\theta) = \sum_{i=1}^n \log p(Y_i; \theta)$. The maximum likelihood estimator is $\hat{\theta} = \arg \max_{\theta} \ell_n(\theta)$. The fitted model for Y is therefore $p(y; \hat{\theta})$. Its fitted mean is

$\hat{\mu} = \mu(\hat{\theta})$. We use $\hat{\mu}$ to predict a future realization Y_{new} .

1.7.3 Prediction-error decomposition

Prediction error is $Y_{\text{new}} - \hat{\mu}$. Insert the true mean:

$$Y_{\text{new}} - \hat{\mu} = \underbrace{Y_{\text{new}} - \mu_0}_{\varepsilon_{\text{new}}} + \underbrace{\mu_0 - \hat{\mu}}_{\text{estimation error}}.$$

Because the future observation is independent of the training sample, $E[\varepsilon_{\text{new}}(\mu_0 - \hat{\mu})] = 0$. Therefore, $E[(Y_{\text{new}} - \hat{\mu})^2] = V_0 + E[(\hat{\mu} - \mu_0)^2]$. Now apply the bias-variance identity: $E[(\hat{\mu} - \mu_0)^2] = \text{Var}(\hat{\mu}) + (E[\hat{\mu}] - \mu_0)^2$. Thus

$$\begin{aligned} E[(Y_{\text{new}} - \hat{\mu})^2] &= \underbrace{V(\theta_0)}_{\text{Irreducible Randomness}} \\ &+ \underbrace{\text{Var}(\mu(\hat{\theta}))}_{\text{Model Variance}} \\ &+ \underbrace{(E[\mu(\hat{\theta})] - \mu(\theta_0))^2}_{\text{Squared Bias}}. \end{aligned}$$

The problem is therefore reduced to estimating these three quantities.

1.7.4 Step 1: Estimate irreducible randomness

The population irreducible error is $R_{\text{irr}} = V(\theta_0)$. Since θ_0 is unknown, use the MLE plug-in estimator: $\hat{R}_{\text{irr}} = V(\hat{\theta})$. Thus the first step is extremely simple: 1. Fit the distribution by MLE. 2. Take the variance formula for that distribution. 3. Replace the unknown parameter by its MLE. For example, if $Y \sim \text{Poisson}(\lambda)$, then $V(\lambda) = \lambda$. Since $\hat{\lambda} = \bar{Y}$, we estimate irreducible randomness by $\hat{R}_{\text{irr}} = \hat{\lambda} = \bar{Y}$.

1.7.5 Step 2: Estimate model variance using Fisher information

The second component is $R_{\text{var}} = \text{Var}(\mu(\hat{\theta}))$. MLE theory gives $\hat{\theta} \stackrel{a}{\sim} N(\theta_0, I_n(\theta_0)^{-1})$, where

$$I_n(\theta) = -E_{\theta} \left[\frac{\partial^2 \ell_n(\theta)}{\partial \theta \partial \theta'} \right]$$

is the Fisher information matrix. Therefore, $\text{Var}(\hat{\theta}) \approx I_n(\theta_0)^{-1}$. If the prediction function $\mu(\theta)$ is nonlinear in θ , apply the delta method: $\mu(\hat{\theta}) \approx \mu(\theta_0) + \nabla \mu(\theta_0)'(\hat{\theta} - \theta_0)$. Hence

$$R_{\text{var}} \approx \nabla \mu(\theta_0)' I_n(\theta_0)^{-1} \nabla \mu(\theta_0).$$

Since θ_0 is unknown, use the plug-in estimator

$$\hat{R}_{\text{var}} = \nabla \mu(\hat{\theta})' I_n(\hat{\theta})^{-1} \nabla \mu(\hat{\theta}).$$

Alternatively, the observed information can be used:

$$J_n(\hat{\theta}) = - \left. \frac{\partial^2 \ell_n(\theta)}{\partial \theta \partial \theta'} \right|_{\theta=\hat{\theta}},$$

giving

$$\hat{R}_{\text{var}} = \nabla \mu(\hat{\theta})' J_n(\hat{\theta})^{-1} \nabla \mu(\hat{\theta}).$$

1.7.6 Step 3: Estimate squared bias

The squared-bias component is $R_{\text{bias}} = (E[\mu(\hat{\theta})] - \mu(\theta_0))^2$. Unlike variance, Fisher information alone does not generally determine finite-sample bias. For regular MLEs, $E[\hat{\theta}] - \theta_0 = O\left(\frac{1}{n}\right)$, so asymptotically $R_{\text{bias}} = O\left(\frac{1}{n^2}\right)$. To first order, MLE theory often approximates $R_{\text{bias}} \approx 0$. Thus the first-order asymptotic prediction decomposition is

$$\widehat{\text{MSE}}_{\text{pred}} \approx V(\hat{\theta}) + \nabla \mu(\hat{\theta})' I_n(\hat{\theta})^{-1} \nabla \mu(\hat{\theta}).$$

However, if a finite-sample bias estimate is desired, a parametric bootstrap provides a practical method.

1.7.7 Parametric bootstrap estimation of model variance and bias

After obtaining the original MLE $\hat{\theta}$, treat $p(y; \hat{\theta})$ as an approximation to the true data-generating distribution. Generate B bootstrap samples: $D^{*(b)} =$

$(Y_1^{*(b)}, \dots, Y_n^{*(b)})$, $b = 1, \dots, B$, with $Y_i^{*(b)} \stackrel{\text{iid}}{\sim} p(y; \hat{\theta})$. For each bootstrap sample calculate the MLE $\hat{\theta}^{*(b)}$ and corresponding mean prediction $\hat{\mu}^{*(b)} = \mu(\hat{\theta}^{*(b)})$. Define $\bar{\mu}^* = \frac{1}{B} \sum_{b=1}^B \hat{\mu}^{*(b)}$. Then estimate model variance by

$$\hat{R}_{\text{var}}^{\text{boot}} = \frac{1}{B-1} \sum_{b=1}^B \left(\hat{\mu}^{*(b)} - \bar{\mu}^* \right)^2.$$

The bootstrap estimates the bias of $\hat{\mu}$ as

$$\widehat{\text{Bias}}_{\text{boot}} = \bar{\mu}^* - \mu(\hat{\theta}).$$

Therefore,

$$\hat{R}_{\text{bias}}^{\text{boot}} = \left(\bar{\mu}^* - \mu(\hat{\theta}) \right)^2.$$

The irreducible component remains $\hat{R}_{\text{irr}} = V(\hat{\theta})$. Thus a completely operational decomposition is

$$\widehat{\text{MSE}}_{\text{pred}} = \hat{R}_{\text{irr}} + \hat{R}_{\text{var}}^{\text{boot}} + \hat{R}_{\text{bias}}^{\text{boot}}.$$

1.7.8 General algorithm

For any correctly specified parametric family $Y \sim p(y; \theta)$, the following procedure can be used.

1. **Specify the parametric distribution.** Determine $p(y; \theta)$, together with $\mu(\theta) = E_{\theta}[Y]$ and $V(\theta) = \text{Var}_{\theta}(Y)$.
2. **Estimate the parameter by MLE.** Compute $\hat{\theta} = \arg \max_{\theta} \sum_{i=1}^n \log p(Y_i; \theta)$.
3. **Construct the fitted predictor.** Use $\hat{\mu} = \mu(\hat{\theta})$.
4. **Estimate irreducible randomness.** Compute $\hat{R}_{\text{irr}} = V(\hat{\theta})$.
5. **Estimate model variance.** Analytically,

$$\hat{R}_{\text{var}} \approx \nabla \mu(\hat{\theta})' I_n(\hat{\theta})^{-1} \nabla \mu(\hat{\theta}).$$

Alternatively, use the parametric-bootstrap variance

$$\hat{R}_{\text{var}}^{\text{boot}} = \text{Var}_* [\mu(\hat{\theta}^*)].$$

6. **Estimate bias.** Estimate $\text{Bias}(\hat{\mu}) = E[\mu(\hat{\theta})] - \mu(\theta_0)$. Using the parametric bootstrap,

$$\widehat{\text{Bias}} = E_* [\mu(\hat{\theta}^*)] - \mu(\hat{\theta}).$$

Then $\hat{R}_{\text{bias}} = \widehat{\text{Bias}}^2$.

7. **Combine the three components.** Finally,

$$\widehat{\text{MSE}}_{\text{pred}} = \hat{R}_{\text{irr}} + \hat{R}_{\text{var}} + \hat{R}_{\text{bias}}.$$

1.7.9 Example 1: Normal distribution with known variance

Suppose $Y_i \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$, where σ^2 is known. The MLE is $\hat{\mu} = \bar{Y}$. The population mean function is simply $m(\mu) = \mu$. The population variance is $V(\mu) = \sigma^2$. Therefore the estimated irreducible error is exactly $\boxed{\hat{R}_{\text{irr}} = \sigma^2}$. The Fisher information in one observation is $I_1(\mu) = \frac{1}{\sigma^2}$. Thus $I_n(\mu) = \frac{n}{\sigma^2}$. Since $\frac{\partial m(\mu)}{\partial \mu} = 1$, model variance is $R_{\text{var}} = I_n(\mu)^{-1} = \frac{\sigma^2}{n}$.

Furthermore, $E[\hat{\mu}] = \mu$, so $R_{\text{bias}} = 0$. Therefore, $\boxed{\text{MSE}_{\text{pred}} = \sigma^2 + \frac{\sigma^2}{n}}$. Thus

$$\boxed{\begin{aligned} R_{\text{irr}} &= \sigma^2, \\ R_{\text{var}} &= \frac{\sigma^2}{n}, \\ R_{\text{bias}} &= 0. \end{aligned}}$$

1.7.10 Example 2: Normal distribution with unknown mean and variance

Suppose $Y_i \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$. Now

$$\theta = \begin{pmatrix} \mu \\ \sigma^2 \end{pmatrix}.$$

The MLEs are $\hat{\mu} = \bar{Y}$ and $\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \bar{Y})^2$. For prediction of a new Y , $\mu(\theta) = \mu$. Thus

$$\nabla \mu(\theta) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The irreducible component is $R_{\text{irr}} = \sigma^2$. A direct plug-in estimate is $\boxed{\hat{R}_{\text{irr}} = \hat{\sigma}_{\text{MLE}}^2}$. The Fisher information matrix is

$$I_n(\theta) = \begin{pmatrix} \frac{n}{\sigma^2} & 0 \\ 0 & \frac{n}{2\sigma^4} \end{pmatrix}.$$

Thus

$$I_n(\theta)^{-1} = \begin{pmatrix} \frac{\sigma^2}{n} & 0 \\ 0 & \frac{2\sigma^4}{n} \end{pmatrix}.$$

Consequently,

$$R_{\text{var}} \approx \nabla \mu(\theta)' I_n(\theta)^{-1} \nabla \mu(\theta) = \frac{\sigma^2}{n}.$$

Its plug-in estimate is $\boxed{\hat{R}_{\text{var}} = \frac{\hat{\sigma}^2}{n}}$. Since $\bar{Y}$ is unbiased for μ , $\boxed{R_{\text{bias}} = 0}$. Notice an important distinction: although the MLE of σ^2 is biased, our prediction target is the mean $\mu(\theta) = \mu$. Therefore the relevant bias for the bias–variance decomposition is the bias of the **prediction function** $\mu(\hat{\theta})$, not necessarily the bias of every individual parameter estimate.

1.7.11 Example 3: Poisson distribution

Suppose $Y_i \stackrel{\text{iid}}{\sim} \text{Poisson}(\lambda)$. The likelihood is $L(\lambda) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{Y_i}}{Y_i!}$. The MLE is $\hat{\lambda} = \bar{Y}$. For the Poisson distribution, $\mu(\lambda) = \lambda$ and $V(\lambda) = \lambda$. Therefore, $\hat{R}_{\text{irr}} = \hat{\lambda}$. The Fisher information in one observation is $I_1(\lambda) = \frac{1}{\lambda}$. Thus $I_n(\lambda) = \frac{n}{\lambda}$. Because $\mu'(\lambda) = 1$, we obtain $R_{\text{var}} = \frac{\lambda}{n}$. Its plug-in estimator is $\hat{R}_{\text{var}} = \frac{\hat{\lambda}}{n}$. Since $E[\hat{\lambda}] = \lambda$, we have $R_{\text{bias}} = 0$. Hence

$$\widehat{\text{MSE}}_{\text{pred}} = \hat{\lambda} + \frac{\hat{\lambda}}{n}.$$

1.7.12 Example 4: Bernoulli distribution

Suppose $Y_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p)$. Then $P(Y_i = 1) = p$, $P(Y_i = 0) = 1 - p$. The MLE is $\hat{p} = \bar{Y}$. The mean and variance are $\mu(p) = p$ and $V(p) = p(1 - p)$. Thus the estimated irreducible randomness is $\hat{R}_{\text{irr}} = \hat{p}(1 - \hat{p})$. The Fisher information for one observation is $I_1(p) = \frac{1}{p(1-p)}$. Therefore, $I_n(p) = \frac{n}{p(1-p)}$. Since $\mu'(p) = 1$, model variance is $R_{\text{var}} = \frac{p(1-p)}{n}$. The plug-in estimate is $\hat{R}_{\text{var}} = \frac{\hat{p}(1 - \hat{p})}{n}$. Because $E[\hat{p}] = p$, the squared bias is $R_{\text{bias}} = 0$.

1.7.13 Example 5: Exponential distribution

Suppose $Y_i \stackrel{\text{iid}}{\sim} \text{Exponential}(\lambda)$, using the rate parameterization $p(y; \lambda) = \lambda e^{-\lambda y}$, $y \geq 0$. The mean and variance are $\mu(\lambda) = \frac{1}{\lambda}$ and $V(\lambda) = \frac{1}{\lambda^2}$. The MLE is $\hat{\lambda} = \frac{1}{\bar{Y}}$. The estimated mean used for prediction is therefore $\mu(\hat{\lambda}) = \frac{1}{\hat{\lambda}} = \bar{Y}$. The estimated irreducible randomness is $\hat{R}_{\text{irr}} = \frac{1}{\hat{\lambda}^2}$. The Fisher information in one observation is $I_1(\lambda) = \frac{1}{\lambda^2}$. Thus $I_n(\lambda) = \frac{n}{\lambda^2}$ and $I_n(\lambda)^{-1} = \frac{\lambda^2}{n}$. Now $\mu'(\lambda) = -\frac{1}{\lambda^2}$. Therefore,

$$\begin{aligned} R_{\text{var}} &\approx \left(-\frac{1}{\lambda^2}\right)^2 \frac{\lambda^2}{n} \\ &= \frac{1}{n\lambda^2}. \end{aligned}$$

Hence, $\hat{R}_{\text{var}} \approx \frac{1}{n\hat{\lambda}^2}$. Interestingly, although $\hat{\lambda} = \frac{1}{\bar{Y}}$ is biased for λ , the prediction of the mean is $\mu(\hat{\lambda}) = \bar{Y}$, and $E[\bar{Y}] = \frac{1}{\lambda}$. Therefore the predictor of the mean is unbiased: $R_{\text{bias}} = 0$. This example demonstrates why the bias relevant for prediction is $E[\mu(\hat{\theta})] - \mu(\theta_0)$, not necessarily $E[\hat{\theta}] - \theta_0$.

1.7.14 A model in which nonlinear transformation can create prediction bias

In general, $E[\hat{\theta}] = \theta_0$ does not imply $E[\mu(\hat{\theta})] = \mu(\theta_0)$ when $\mu(\theta)$ is nonlinear. Using a second-order Taylor expansion,

$$\mu(\hat{\theta}) \approx \mu(\theta_0) + \nabla\mu(\theta_0)'(\hat{\theta} - \theta_0) + \frac{1}{2}(\hat{\theta} - \theta_0)'H_\mu(\theta_0)(\hat{\theta} - \theta_0),$$

where $H_\mu(\theta_0)$ is the Hessian matrix of $\mu(\theta)$. Taking expectations gives approximately

$$\begin{aligned} E[\mu(\hat{\theta})] - \mu(\theta_0) &\approx \nabla\mu(\theta_0)' \text{Bias}(\hat{\theta}) \\ &\quad + \frac{1}{2} \text{tr} [H_\mu(\theta_0) \text{Var}(\hat{\theta})]. \end{aligned}$$

Using Fisher information,

$$\text{Bias}(\mu(\hat{\theta})) \approx \nabla\mu(\theta_0)' \text{Bias}(\hat{\theta}) + \frac{1}{2} \text{tr} [H_\mu(\theta_0) I_n(\theta_0)^{-1}].$$

Thus even when $\text{Bias}(\hat{\theta}) = 0$, curvature of the prediction function $\mu(\theta)$ can produce finite-sample prediction bias. This is the second-order analogue of the delta method.

1.7.14.1 Example: Exponential distribution and prediction bias

Consider $Y_i \stackrel{\text{iid}}{\sim} \text{Exponential}(\lambda)$, with $E[Y_i] = \frac{1}{\lambda}$, $\text{Var}(Y_i) = \frac{1}{\lambda^2}$. The MLE of the rate parameter is $\hat{\lambda} = \frac{1}{\bar{Y}}$. The quantity we ultimately use for prediction, however, is the mean of the distribution: $\mu(\lambda) = \frac{1}{\lambda}$. Therefore the fitted mean is $\mu(\hat{\lambda}) = \frac{1}{\hat{\lambda}} = \bar{Y}$. We know exactly that $E[\bar{Y}] = \frac{1}{\lambda}$, so $E[\mu(\hat{\lambda})] = \mu(\lambda)$. Consequently, the fitted mean has zero prediction bias: $\text{Bias}(\mu(\hat{\lambda})) = 0$. At the same time, however, $\hat{\lambda} = \frac{1}{\bar{Y}}$ is itself biased. The second-order Taylor formula explains how these two statements can simultaneously be true.

Applying the second-order bias formula

For a scalar parameter, the general approximation

$$\text{Bias}(\mu(\hat{\theta})) \approx \nabla\mu(\theta_0)' \text{Bias}(\hat{\theta}) + \frac{1}{2} \text{tr} [H_\mu(\theta_0) \text{Var}(\hat{\theta})]$$

reduces to

$$\text{Bias}(\mu(\hat{\theta})) \approx \mu'(\theta_0) \text{Bias}(\hat{\theta}) + \frac{1}{2} \mu''(\theta_0) \text{Var}(\hat{\theta}).$$

For the Exponential model, $\theta = \lambda$ and $\mu(\lambda) = \frac{1}{\lambda}$. Therefore, $\mu'(\lambda) = -\frac{1}{\lambda^2}$ and $\mu''(\lambda) = \frac{2}{\lambda^3}$. Hence

$$\text{Bias}(\mu(\hat{\lambda})) \approx -\frac{1}{\lambda^2} \text{Bias}(\hat{\lambda}) + \frac{1}{\lambda^3} \text{Var}(\hat{\lambda}).$$

We now evaluate these two terms.

Step 1: Approximate the bias of $\hat{\lambda}$

We know exactly that $E[\hat{\lambda}] = \frac{n}{n-1}\lambda$. Therefore, $\text{Bias}(\hat{\lambda}) = \frac{\lambda}{n-1}$. For large n , $\frac{1}{n-1} \approx \frac{1}{n}$, so to first order, $\boxed{\text{Bias}(\hat{\lambda}) \approx \frac{\lambda}{n}}$. The contribution of parameter bias to prediction bias is therefore

$$\mu'(\lambda) \text{Bias}(\hat{\lambda}) \approx -\frac{1}{\lambda^2} \frac{\lambda}{n}.$$

Thus $\boxed{\mu'(\lambda) \text{Bias}(\hat{\lambda}) \approx -\frac{1}{n\lambda}}$. If this were the only term, we would conclude that prediction of the mean was downward biased. But there is a second term.

Step 2: Account for the curvature of $\mu(\lambda)$

Maximum likelihood theory gives $\text{Var}(\hat{\lambda}) \approx I_n(\lambda)^{-1}$. For an Exponential distribution, $I_n(\lambda) = \frac{n}{\lambda^2}$, so $\boxed{\text{Var}(\hat{\lambda}) \approx \frac{\lambda^2}{n}}$. The curvature contribution is therefore $\frac{1}{2}\mu''(\lambda) \text{Var}(\hat{\lambda})$. Substituting,

$$\begin{aligned} \frac{1}{2}\mu''(\lambda) \text{Var}(\hat{\lambda}) &\approx \frac{1}{2} \left(\frac{2}{\lambda^3} \right) \left(\frac{\lambda^2}{n} \right) \\ &= \frac{1}{n\lambda}. \end{aligned}$$

Hence,

$$\boxed{\frac{1}{2}\mu''(\lambda) \text{Var}(\hat{\lambda}) \approx +\frac{1}{n\lambda}}.$$

Step 3: The two effects cancel

Putting the two terms together,

$$\begin{aligned} \text{Bias}(\mu(\hat{\lambda})) &\approx \mu'(\lambda) \text{Bias}(\hat{\lambda}) + \frac{1}{2}\mu''(\lambda) \text{Var}(\hat{\lambda}) \\ &\approx -\frac{1}{n\lambda} + \frac{1}{n\lambda} \\ &= 0. \end{aligned}$$

Therefore, $\boxed{\text{Bias}(\mu(\hat{\lambda})) \approx 0}$. In this particular example the result is actually stronger than the Taylor approximation suggests. We know exactly that $\mu(\hat{\lambda}) = \frac{1}{\hat{\lambda}} = \bar{Y}$, and therefore $E[\mu(\hat{\lambda})] = E[\bar{Y}] = \frac{1}{\lambda} = \mu(\lambda)$. Hence $\boxed{\text{Bias}(\mu(\hat{\lambda})) = 0}$ exactly.

Why is $\hat{\lambda}$ biased in the first place?

There is also a useful way to see the same curvature phenomenon one step earlier. We start with the unbiased estimator $\bar{Y}$ of $\frac{1}{\lambda}$. But the rate estimator is obtained through the nonlinear transformation $g(x) = \frac{1}{x}$. Thus $\hat{\lambda} = g(\bar{Y})$. Even though $E[\bar{Y}] = \frac{1}{\lambda}$, we generally have $E[g(\bar{Y})] \neq g(E[\bar{Y}])$. Using a second-order Taylor expansion around $m = E[\bar{Y}] = \frac{1}{\lambda}$, we have $E[g(\bar{Y})] \approx g(m) + \frac{1}{2}g''(m) \text{Var}(\bar{Y})$. For $g(x) = \frac{1}{x}$, we have $g''(x) = \frac{2}{x^3}$. At $m = \frac{1}{\lambda}$,

this becomes $g''(m) = 2\lambda^3$. Also, $\text{Var}(\bar{Y}) = \frac{1}{n\lambda^2}$. Therefore,

$$\begin{aligned} E[\hat{\lambda}] &= E[g(\bar{Y})] \\ &\approx g\left(\frac{1}{\lambda}\right) + \frac{1}{2}(2\lambda^3)\frac{1}{n\lambda^2} \\ &= \lambda + \frac{\lambda}{n}. \end{aligned}$$

Hence $\text{Bias}(\hat{\lambda}) \approx \frac{\lambda}{n}$. This agrees with the first-order expansion of the exact bias $\text{Bias}(\hat{\lambda}) = \frac{\lambda}{n-1} \approx \frac{\lambda}{n}$. Thus the bias in $\hat{\lambda}$ itself comes directly from the curvature of the reciprocal transformation.

The full picture

There are therefore two nonlinear transformations in this example. First, $\bar{Y} \xrightarrow{g(x)=1/x} \hat{\lambda}$. Because g is nonlinear and convex, $\bar{Y}$ unbiased $\not\Rightarrow \hat{\lambda}$ unbiased.

Indeed, $\text{Bias}(\hat{\lambda}) > 0$. Then prediction applies the mean function $\hat{\lambda} \xrightarrow{\mu(\lambda)=1/\lambda} \mu(\hat{\lambda})$. But these reciprocal transformations undo each other: $\mu(\hat{\lambda}) = \frac{1}{1/\bar{Y}} = \bar{Y}$. Consequently,

$\hat{\lambda}$ is biased, $\mu(\hat{\lambda})$ is unbiased. The second-order formula reveals how this happens:

$$\underbrace{\mu'(\lambda) \text{Bias}(\hat{\lambda})}_{-\frac{1}{n\lambda}} + \underbrace{\frac{1}{2}\mu''(\lambda) \text{Var}(\hat{\lambda})}_{+\frac{1}{n\lambda}} \approx 0.$$

Thus the prediction bias depends not simply on whether the MLE itself is biased, but on the interaction among

parameter bias + curvature of the prediction function + parameter variance.

This is precisely why the relevant bias term in the prediction bias–variance decomposition is $E[\mu(\hat{\theta})] - \mu(\theta_0)$, rather than $E[\hat{\theta}] - \theta_0$.

1.7.14.2 Example: Lognormal distribution and prediction bias

A very clean example is the **lognormal distribution**. It is actually better than the exponential example for demonstrating the second-order formula because the mean is a genuinely nonlinear function of the parameters, and the nonlinearity creates prediction bias. Suppose $Y_i \sim \text{Lognormal}(\mu, \sigma^2)$, meaning $\log Y_i \sim N(\mu, \sigma^2)$. The mean of Y is

$m(\mu, \sigma^2) = E[Y] = e^{\mu + \sigma^2/2}$. This is nonlinear in **both parameters**.

Start with the simpler case: σ^2 known

Suppose σ^2 is known and only μ is estimated. The MLE is $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n \log Y_i$. Since $\log Y_i \sim N(\mu, \sigma^2)$, we know exactly that $E[\hat{\mu}] = \mu$ and $\text{Var}(\hat{\mu}) = \frac{\sigma^2}{n}$. Thus, $\text{Bias}(\hat{\mu}) = 0$. If our objective is to predict the mean of Y , however, the fitted mean is $\hat{m} = m(\hat{\mu}) = e^{\hat{\mu} + \sigma^2/2}$. Although $\hat{\mu}$ is unbiased, $\hat{m}$ is **not** unbiased because the exponential transformation is nonlinear. The second-order formula shows exactly why. For $m(\mu) = e^{\mu + \sigma^2/2}$, the first

and second derivatives are $m'(\mu) = e^{\mu+\sigma^2/2} = m(\mu)$ and $m''(\mu) = e^{\mu+\sigma^2/2} = m(\mu)$. The second-order bias approximation is

$$\text{Bias}(m(\hat{\mu})) \approx m'(\mu) \text{Bias}(\hat{\mu}) + \frac{1}{2}m''(\mu) \text{Var}(\hat{\mu}).$$

Since $\text{Bias}(\hat{\mu}) = 0$, the first term disappears: $\text{Bias}(m(\hat{\mu})) \approx \frac{1}{2}m''(\mu) \text{Var}(\hat{\mu})$. Substitute the expressions above: $\text{Bias}(m(\hat{\mu})) \approx \frac{1}{2}e^{\mu+\sigma^2/2} \frac{\sigma^2}{n}$. Therefore, $\text{Bias}(m(\hat{\mu})) \approx \frac{\sigma^2}{2n} e^{\mu+\sigma^2/2}$.

Since $m(\mu) = e^{\mu+\sigma^2/2}$, we can write this particularly neatly as $\text{Bias}(\hat{m}) \approx m(\mu) \frac{\sigma^2}{2n}$. This is exactly the phenomenon this example is meant to illustrate:

$$\boxed{\text{Bias}(\hat{\mu}) = 0 \not\Rightarrow \text{Bias}(m(\hat{\mu})) = 0.}$$

The reason is the curvature term $\frac{1}{2}m''(\mu) \text{Var}(\hat{\mu})$.

We can even verify the approximation exactly

This example is especially nice because the exact bias is available. We know $\hat{\mu} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$. Therefore, $E[e^{\hat{\mu}}] = e^{\mu+\frac{1}{2}\sigma^2/n}$. Hence

$$\begin{aligned} E[\hat{m}] &= E\left[e^{\hat{\mu}+\sigma^2/2}\right] \\ &= e^{\sigma^2/2} E[e^{\hat{\mu}}] \\ &= e^{\mu+\sigma^2/2} e^{\sigma^2/(2n)}. \end{aligned}$$

Since $m(\mu) = e^{\mu+\sigma^2/2}$, the exact bias is $\text{Bias}(\hat{m}) = m(\mu) \left[e^{\sigma^2/(2n)} - 1\right]$. Now use $e^x =$

$1+x+\frac{x^2}{2}+\dots$. With $x = \frac{\sigma^2}{2n}$, we obtain $e^{\sigma^2/(2n)} - 1 \approx \frac{\sigma^2}{2n}$. Therefore, $\text{Bias}(\hat{m}) \approx m(\mu) \frac{\sigma^2}{2n}$, which is precisely what the second-order Taylor formula gave us.

Why this is a stronger teaching example

The exponential example had $\hat{\lambda}$ biased but $m(\hat{\lambda})$ unbiased, because the reciprocal transformations happened to cancel. The lognormal example gives the complementary and perhaps more important phenomenon: $\hat{\mu}$ unbiased but $m(\hat{\mu})$ biased. The source of the bias is entirely the **curvature of the mean function**: $m''(\mu) > 0$. So this example directly demonstrates the purpose of the second-order term in the formula:

$$\boxed{\text{Bias}(m(\hat{\theta})) \approx m'(\theta) \text{Bias}(\hat{\theta}) + \frac{1}{2}m''(\theta) \text{Var}(\hat{\theta}).}$$

Even when the first term is zero, the second need not be.

1.7.14.3 Example: Lognormal distribution with both parameters unknown

Then the example becomes more interesting because **both parameters are estimated**, and both contribute to the bias and variance of the estimated mean. Let

$Y_i \sim \text{Lognormal}(\mu, \sigma^2)$, so $Z_i = \log Y_i \sim N(\mu, \sigma^2)$. The population mean of the original Y is $m(\mu, \sigma^2) = e^{\mu + \sigma^2/2}$. Let us use the parameter vector

$$\theta = \begin{pmatrix} \mu \\ v \end{pmatrix}, \quad v = \sigma^2.$$

MLEs of both parameters

Since $Z_i = \log Y_i$ is normal, $\hat{\mu} = \bar{Z}$ and $\hat{v} = \hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum_{i=1}^n (Z_i - \bar{Z})^2$. Therefore the natural MLE plug-in estimate of the mean of Y is $\hat{m} = m(\hat{\mu}, \hat{v}) = e^{\hat{\mu} + \hat{v}/2}$. Unlike the known- σ^2 case, both $\hat{\mu}$ and $\hat{v}$ are random. Moreover, $E[\hat{\mu}] = \mu$, but $E[\hat{v}] = \frac{n-1}{n}v$. Therefore, $\text{Bias}(\hat{\mu}) = 0$, $\text{Bias}(\hat{v}) = -\frac{v}{n}$. So now we have **two sources of prediction bias**: bias in the estimated variance parameter and curvature of $m(\mu, v)$.

Apply the multivariate second-order formula

The general formula is

$$\text{Bias}(m(\hat{\theta})) \approx \nabla m(\theta)' \text{Bias}(\hat{\theta}) + \frac{1}{2} \text{tr} [H_m(\theta) \text{Var}(\hat{\theta})].$$

We calculate each component. Since $m(\mu, v) = e^{\mu + v/2}$, the gradient is $\frac{\partial m}{\partial \mu} = m$, $\frac{\partial m}{\partial v} = \frac{1}{2}m$. Thus

$$\nabla m(\theta) = m \begin{pmatrix} 1 \\ 1/2 \end{pmatrix}.$$

The Hessian is

$$H_m(\theta) = m \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1/4 \end{pmatrix}.$$

Bias contribution from the MLE parameters

The bias vector is

$$\text{Bias}(\hat{\theta}) = \begin{pmatrix} 0 \\ -v/n \end{pmatrix}.$$

Therefore,

$$\begin{aligned} \nabla m(\theta)' \text{Bias}(\hat{\theta}) &= m \begin{pmatrix} 1 & 1/2 \end{pmatrix} \begin{pmatrix} 0 \\ -v/n \end{pmatrix} \\ &= -\frac{mv}{2n}. \end{aligned}$$

Hence the direct parameter-bias contribution is $-\frac{mv}{2n}$. This is negative because the MLE of $v = \sigma^2$ is downward biased.

Curvature contribution

For the normal model, asymptotically, $\text{Var}(\hat{\mu}) \approx \frac{v}{n}$, and $\text{Var}(\hat{v}) \approx \frac{2v^2}{n}$. Also, $\hat{\mu}$ and $\hat{v}$ are independent in the normal model, so $\text{Cov}(\hat{\mu}, \hat{v}) = 0$. Thus

$$\text{Var}(\hat{\theta}) \approx \begin{pmatrix} v/n & 0 \\ 0 & 2v^2/n \end{pmatrix}.$$

Now calculate $\frac{1}{2} \text{tr} [H_m(\theta) \text{Var}(\hat{\theta})]$. Substituting,

$$\begin{aligned} & \frac{1}{2} \text{tr} \left[m \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1/4 \end{pmatrix} \begin{pmatrix} v/n & 0 \\ 0 & 2v^2/n \end{pmatrix} \right] \\ &= \frac{1}{2} m \left[\frac{v}{n} + \frac{1}{4} \frac{2v^2}{n} \right]. \end{aligned}$$

Therefore, $\text{Curvature contribution} = \frac{mv}{2n} + \frac{mv^2}{4n}.$

Something interesting happens

Combine the parameter-bias term and curvature term: $\text{Bias}(\hat{m}) \approx -\frac{mv}{2n} + \frac{mv}{2n} + \frac{mv^2}{4n}$. The

first two terms cancel: $-\frac{mv}{2n} + \frac{mv}{2n} = 0$. Thus, $\text{Bias}(\hat{m}) \approx \frac{mv^2}{4n}.$ Since $v = \sigma^2$, $v^2 = \sigma^4$,

so $\text{Bias}(\hat{m}) \approx \frac{m\sigma^4}{4n}.$ Or, since $m = e^{\mu+\sigma^2/2}$, $\text{Bias}(\hat{m}) \approx \frac{\sigma^4}{4n} e^{\mu+\sigma^2/2}.$ So the plug-in MLE estimator $e^{\hat{\mu}+\hat{\sigma}^2/2}$ is approximately **upward biased** for the population mean.

What about model variance?

The same gradient and Fisher-information covariance matrix give the first-order model variance:

$$\text{Var}(\hat{m}) \approx \nabla m(\theta)' \text{Var}(\hat{\theta}) \nabla m(\theta).$$

Thus

$$\begin{aligned} \text{Var}(\hat{m}) &\approx m^2 \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1/4 \end{pmatrix} \begin{pmatrix} v/n & 0 \\ 0 & 2v^2/n \end{pmatrix} \begin{pmatrix} 1 \\ 1/2 \end{pmatrix} \\ &= m^2 \left(\frac{v}{n} + \frac{v^2}{2n} \right). \end{aligned}$$

Therefore,

$$\text{R}_{\text{var}} \approx \frac{m^2}{n} \left(\sigma^2 + \frac{\sigma^4}{2} \right).$$

And irreducible randomness?

For a lognormal random variable, $\text{Var}(Y) = (e^{\sigma^2} - 1)e^{2\mu+\sigma^2}$. Since $m^2 = e^{2\mu+\sigma^2}$, we can write $R_{\text{irr}} = m^2(e^{\sigma^2} - 1).$ The plug-in estimate is therefore

$$\hat{R}_{\text{irr}} = e^{2\hat{\mu}+\hat{\sigma}^2} (e^{\hat{\sigma}^2} - 1).$$

So for the unknown- μ , unknown- σ^2 lognormal model, the three components are approximately

$$\begin{aligned} R_{\text{irr}} &= m^2(e^{\sigma^2} - 1), \\ R_{\text{var}} &\approx \frac{m^2}{n} \left(\sigma^2 + \frac{\sigma^4}{2} \right), \\ R_{\text{bias}} &\approx \left(\frac{m\sigma^4}{4n} \right)^2. \end{aligned}$$

One subtle point is worth emphasizing for the lecture: **bias itself is $O(1/n)$, but squared bias is $O(1/n^2)$** . Thus, for a regular MLE, model variance is generally a first-order $1/n$ contribution to prediction MSE, whereas squared bias is a smaller second-order $1/n^2$ contribution. This is why first-order asymptotic MLE theory often ignores squared bias, even though the second-order calculation is conceptually very useful.

1.7.15 Two practical approaches

There are therefore two useful approaches to estimating the three components.

1.7.16 Approach A: Analytic asymptotic decomposition

Estimate $\hat{R}_{\text{irr}} = V(\hat{\theta})$ and

$$\hat{R}_{\text{var}} = \nabla \mu(\hat{\theta})' I_n(\hat{\theta})^{-1} \nabla \mu(\hat{\theta}).$$

Approximate $\hat{R}_{\text{bias}} \approx 0$ to first order, or use a higher-order MLE bias approximation. This approach makes the role of Fisher information transparent.

1.7.17 Approach B: Parametric-bootstrap decomposition

Estimate $\hat{R}_{\text{irr}} = V(\hat{\theta})$. Generate bootstrap datasets from $p(y; \hat{\theta})$. Re-estimate the model on every bootstrap dataset. Use

$$\hat{R}_{\text{var}} = \text{Var}_* (\mu(\hat{\theta}^*))$$

and

$$\hat{R}_{\text{bias}} = \left[E_* (\mu(\hat{\theta}^*)) - \mu(\hat{\theta}) \right]^2.$$

This approach can capture finite-sample nonlinearities that are omitted by first-order Fisher-information approximations.

1.7.18 Final summary

For a parametric model $Y \sim p(y; \theta_0)$, define $\mu(\theta) = E_\theta[Y]$ and $V(\theta) = \text{Var}_\theta(Y)$. Fit the model by MLE: $\hat{\theta} = \arg \max_\theta \ell_n(\theta)$. The prediction of a new observation is $\hat{Y} = \mu(\hat{\theta})$.

The fundamental decomposition is

$$E[(Y_{\text{new}} - \mu(\hat{\theta}))^2] = \underbrace{V(\theta_0)}_{\text{irreducible randomness}} + \underbrace{\text{Var}(\mu(\hat{\theta}))}_{\text{model variance}} + \underbrace{[E(\mu(\hat{\theta})) - \mu(\theta_0)]^2}_{\text{squared bias}}.$$

The operational estimates are $\hat{R}_{\text{irr}} = V(\hat{\theta})$ and, using Fisher information,

$$\hat{R}_{\text{var}} \approx \nabla \mu(\hat{\theta})' I_n(\hat{\theta})^{-1} \nabla \mu(\hat{\theta}).$$

For squared bias, either use the first-order approximation $\hat{R}_{\text{bias}} \approx 0$ for a regular MLE in a sufficiently large sample, or estimate it by a parametric bootstrap:

$$\hat{R}_{\text{bias}} = [E_*(\mu(\hat{\theta}^*)) - \mu(\hat{\theta})]^2.$$

The conceptual roles are therefore

$$\begin{array}{ll} V(\theta_0) & \longrightarrow \text{randomness of the future } Y, \\ I_n(\theta_0)^{-1} & \longrightarrow \text{randomness from estimating } \theta_0, \\ E[\mu(\hat{\theta})] - \mu(\theta_0) & \longrightarrow \text{systematic estimation error.} \end{array}$$

The first persists even with infinite data. The second and, under regularity, the third disappear as the sample size becomes large.

1.8 Where does higher-order MLE bias come from?

Motivating question: bias for the MLE is either 0 or $O(1/n)$ in the examples so far — can it be some other order, and where does the $1/n$ term come from in general? The schematic expansion comes directly from taking the MLE score equation and doing a higher-order Taylor expansion around the true parameter. The distinction between Z and B becomes quite natural once we derive it. For clarity, consider a scalar parameter θ .

1.8.1 Understanding approximation to score equation

1.8.1.1 Start from the score equation

For i.i.d. observations, $\ell_n(\theta) = \sum_{i=1}^n \ell_i(\theta)$, $\ell_i(\theta) = \log f(Y_i; \theta)$. Define the score $S_n(\theta) = \ell'_n(\theta) = \sum_{i=1}^n s_i(\theta)$. The MLE satisfies $S_n(\hat{\theta}) = 0$. We want to understand the random difference $\Delta = \hat{\theta} - \theta_0$.

1.8.1.2 Taylor-expand the score around the true parameter

Expand $S_n(\hat{\theta})$ around θ_0 : $0 = S_n(\theta_0) + S'_n(\theta_0)\Delta + \frac{1}{2}S''_n(\theta_0)\Delta^2 + \dots$. That is,

$$0 = S_n + H_n\Delta + \frac{1}{2}T_n\Delta^2 + \dots \quad \text{where } S_n = S_n(\theta_0), \quad H_n = \ell''_n(\theta_0), \quad T_n = \ell'''_n(\theta_0).$$

The first-order approximation simply ignores the quadratic and higher terms: $0 \approx S_n + H_n\Delta$. Therefore, $\Delta \approx -H_n^{-1}S_n$. This is the usual first-order MLE approximation.

1.8.1.3 What are the orders of the score and Hessian?

This is the key. The score is a sum: $S_n = \sum_{i=1}^n s_i(\theta_0)$. Under the regular likelihood conditions, $E[s_i(\theta_0)] = 0$. Why? Because

$$\begin{aligned} E_{\theta_0}[s_i(\theta_0)] &= \int \frac{\partial}{\partial \theta} \log f(y; \theta_0) f(y; \theta_0) dy \\ &= \int \frac{\partial f(y; \theta_0) / \partial \theta}{f(y; \theta_0)} f(y; \theta_0) dy \\ &= \int \frac{\partial f(y; \theta_0)}{\partial \theta} dy \\ &= \frac{\partial}{\partial \theta} \int f(y; \theta_0) dy \\ &= \frac{\partial}{\partial \theta} 1 \\ &= 0. \end{aligned}$$

So each score contribution has mean zero. By the central limit theorem,

$$\frac{1}{\sqrt{n}}S_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n s_i(\theta_0) \xrightarrow{d} N(0, I(\theta_0)).$$

Thus $S_n = O_p(\sqrt{n})$. By contrast, the Hessian is also a sum: $H_n = \sum_{i=1}^n \ell'''_i(\theta_0)$. But its individual terms generally have a nonzero mean: $E[\ell'''_i(\theta_0)] = -I(\theta_0)$. Hence, by the law of large numbers, $\frac{1}{n}H_n \xrightarrow{p} -I(\theta_0)$, so $H_n = O_p(n)$. Therefore, $\Delta \approx -H_n^{-1}S_n$ has order $O_p(1/n) \times O_p(\sqrt{n}) = O_p(1/\sqrt{n})$. This is where $\hat{\theta} - \theta_0 = O_p(1/\sqrt{n})$ comes from.

1.8.1.4 Where does the $Z/\sqrt{n}$ term come from?

More explicitly, $H_n \approx -nI(\theta_0)$. Therefore, $\hat{\theta} - \theta_0 \approx \frac{1}{nI(\theta_0)}S_n$. Rewrite this as

$$\hat{\theta} - \theta_0 \approx \frac{1}{\sqrt{n}} \left[I(\theta_0)^{-1} \frac{S_n}{\sqrt{n}} \right].$$

Define $Z_n = I(\theta_0)^{-1} \frac{S_n}{\sqrt{n}}$. Then $\hat{\theta} - \theta_0 \approx \frac{1}{\sqrt{n}}Z_n$. And asymptotically, $Z_n \xrightarrow{d} Z$, $Z \sim$

$N(0, I(\theta_0)^{-1})$. So $E[Z] = 0$. This is not accidental. It comes from the fundamental likelihood identity $E[S_n(\theta_0)] = 0$. The first-order estimation error is essentially a scaled score, and the score is mean zero at the true parameter.

1.8.1.5 But this alone would suggest zero bias

Exactly. If $\hat{\theta} - \theta_0 = \frac{1}{\sqrt{n}}Z$ were the whole story, then $E[\hat{\theta} - \theta_0] = \frac{1}{\sqrt{n}}E[Z] = 0$. So a first-order asymptotic approximation cannot explain the usual finite-sample bias of an MLE. We must go to second order.

1.8.1.6 The second-order terms are nonlinear

Return to $0 = S_n + H_n\Delta + \frac{1}{2}T_n\Delta^2 + \dots$. Solving approximately gives $\Delta \approx -H_n^{-1}S_n - \frac{1}{2}H_n^{-1}T_n(H_n^{-1}S_n)^2 + \dots$, although a complete second-order expansion must also account for the random fluctuation of H_n around its expectation. Look at the order of the curvature term. We have $H_n^{-1} = O_p(1/n)$, $T_n = O_p(n)$, and $H_n^{-1}S_n = O_p(1/\sqrt{n})$. Therefore, $H_n^{-1}T_n(H_n^{-1}S_n)^2$ has order $\frac{1}{n} \times n \times \frac{1}{n} = \boxed{\frac{1}{n}}$. That is the origin of the next-order $1/n$ term.

1.8.1.7 Why can that term have nonzero expectation?

Because it contains squares and products of random quantities. This is the central intuition. The first-order term contains something proportional to S_n , and $E[S_n] = 0$. But the second-order term contains quantities resembling S_n^2 . Even though $E[S_n] = 0$, we certainly do not have $E[S_n^2] = 0$. Instead, $E[S_n^2] = \text{Var}(S_n) = nI(\theta_0)$. So nonlinear transformations of a mean-zero score can have nonzero means. This is the same mathematical phenomenon we saw earlier with nonlinear transformations: $E[Z] = 0$ does not imply $E[Z^2] = 0$. In fact, $E[Z^2] = \text{Var}(Z) > 0$.

1.8.1.8 A simple schematic derivation

Suppose, only schematically, that the likelihood expansion gives $\hat{\theta} - \theta_0 \approx a\frac{S_n}{n} + b\frac{S_n^2}{n^2}$. Because $S_n = O_p(\sqrt{n})$, the first term has order $\sqrt{n}/n = 1/\sqrt{n}$. The second has order $n/n^2 = 1/n$. Now take expectations. For the first term, $E\left[a\frac{S_n}{n}\right] = \frac{a}{n}E[S_n] = 0$. For the second, $E\left[b\frac{S_n^2}{n^2}\right] = \frac{b}{n^2}E[S_n^2]$. Since $E[S_n^2] = nI(\theta_0)$, we obtain $E\left[b\frac{S_n^2}{n^2}\right] = \frac{bI(\theta_0)}{n}$. Therefore,

$$\boxed{E[\hat{\theta} - \theta_0] \approx \frac{B(\theta_0)}{n}}. \text{ This is the essence of second-order MLE bias.}$$

1.8.1.9 One correction to the earlier schematic notation

When $\hat{\theta} - \theta_0$ is written as $\frac{1}{\sqrt{n}}Z + \frac{1}{n}B + \text{smaller terms}$, calling B simply a "systematic bias" is a little too loose. A more accurate expansion is something like

$$\boxed{\hat{\theta} - \theta_0 = \frac{1}{\sqrt{n}}Z_n + \frac{1}{n}W_n + o_p(1/n)}, \text{ where } E[Z_n] \rightarrow 0 \text{ but } W_n \text{ can itself be random and}$$

can have a nonzero expectation: $E[W_n] \rightarrow B(\theta_0) \neq 0$. Then $\text{Bias}(\hat{\theta}) = E[\hat{\theta} - \theta_0] \approx \frac{B(\theta_0)}{n}$.

So it is really $\boxed{B(\theta_0) = \text{mean of the second-order correction,}}$ not necessarily that the entire second-order term is deterministic.

1.8.1.10 The conceptual chain

The entire argument can be summarized as $\text{MLE solves } S_n(\hat{\theta}) = 0$ Taylor-expand the score around θ_0 :

$$0 = S_n(\theta_0) + H_n(\theta_0)(\hat{\theta} - \theta_0) + \frac{1}{2}T_n(\theta_0)(\hat{\theta} - \theta_0)^2 + \dots$$

First-order solution:

$$\hat{\theta} - \theta_0 \sim \underbrace{\frac{1}{\sqrt{n}}Z}_{\text{score-driven random error}}, \quad E[Z] = 0$$

because $E[S_n(\theta_0)] = 0$. Second-order solution introduces terms involving S_n^2 , $S_n H_n$, $T_n S_n^2$, etc., whose expectations generally do not vanish. Therefore,

$$\hat{\theta} - \theta_0 = O_p(n^{-1/2}) + O_p(n^{-1}) + \dots \quad \text{but after taking expectations, } \text{Bias}(\hat{\theta}) = 0 + O(n^{-1}) + \dots$$

So the distinction is: the dominant estimation error is $1/\sqrt{n}$, but it oscillates around zero; the smaller $1/n$ curvature correction need not oscillate around zero and therefore becomes the leading bias. This is also closely related to why the second-order delta-method term $\frac{1}{2}g''(\theta) \text{Var}(\hat{\theta})$ creates $O(1/n)$ bias: both phenomena arise from curvature acting on $O(1/\sqrt{n})$ random fluctuations, whose squares are $O(1/n)$ and have positive/nonzero expectation.

1.8.2 Geometric interpretation of the score equation and its root

Motivating question: how does the second-order solution to Δ come to contain $S_n(\theta_0)^2$, and what is the geometric picture behind approximating the score equation to quadratic order? We have a nonlinear score function $S_n(\theta)$, and the MLE $\hat{\theta}$ is the point where that function crosses zero. We do not know that root explicitly, so we approximate the score near the true parameter θ_0 , and then solve the approximating equation for $\hat{\theta} - \theta_0$. The subtle part is that when we use a quadratic approximation to the score, solving it to second order introduces $S_n(\theta_0)^2$.

1.8.2.1 Geometrically, what are we trying to find?

Consider a scalar parameter. The log-likelihood is $\ell_n(\theta)$. The score is its slope: $S_n(\theta) = \ell'_n(\theta)$. The MLE satisfies $S_n(\hat{\theta}) = 0$. So geometrically, if we plot $S_n(\theta)$ against θ , the MLE is the zero crossing of the score curve. Picture the true parameter θ_0 as a fixed point on the horizontal axis; the score curve is generally nonlinear, and we know its value, slope, and curvature locally around θ_0 , and we want to approximate where it crosses the horizontal axis. The important distinction is $\theta_0 \neq \hat{\theta}$ in a particular sample. At the true population parameter, $E[S_n(\theta_0)] = 0$, but generally $S_n(\theta_0) \neq 0$ in a realized sample. That random displacement of the score from zero is what pushes the sample MLE away from θ_0 .

1.8.2.2 Linear approximation: replace the curve by its tangent

Let $\Delta = \hat{\theta} - \theta_0$. A first-order Taylor expansion gives $S_n(\hat{\theta}) \approx S_n(\theta_0) + S'_n(\theta_0)\Delta$. Since the MLE satisfies $S_n(\hat{\theta}) = 0$, $0 \approx S_n(\theta_0) + S'_n(\theta_0)\Delta$. Solve for Δ : $\Delta \approx -\frac{S_n(\theta_0)}{S'_n(\theta_0)}$.

Geometrically, this says: draw the tangent line to the score curve at θ_0 ; where does that tangent line cross zero? That zero crossing is our first-order approximation to $\hat{\theta}$. This is essentially one Newton step starting at θ_0 .

1.8.2.3 Why does this produce a term proportional to the score?

Let $S = S_n(\theta_0)$ and $H = S'_n(\theta_0)$. Then $\Delta_1 = -S/H$. Approximately, $H \approx -nI(\theta_0)$. Therefore, $\Delta_1 \approx \frac{S}{nI(\theta_0)}$. Since $S = O_p(\sqrt{n})$, we get $\Delta_1 = O_p(n^{-1/2})$. Most importantly, $E[S] = 0$. Thus this first-order movement of the root is centered around zero.

1.8.2.4 Now suppose the score curve is curved

A tangent line ignores curvature. Use a second-order Taylor approximation:

$$S_n(\hat{\theta}) \approx S_n(\theta_0) + S'_n(\theta_0)\Delta + \frac{1}{2}S''_n(\theta_0)\Delta^2.$$

Define $S = S_n(\theta_0)$, $H = S'_n(\theta_0)$, $T = S''_n(\theta_0)$. Then $0 \approx S + H\Delta + \frac{1}{2}T\Delta^2$. And yes: this is precisely a quadratic approximation to the nonlinear score equation in terms of $\Delta = \hat{\theta} - \theta_0$. We then want to find the root of this approximating parabola.

1.8.2.5 Why don't we simply use the quadratic formula?

We could. From $S + H\Delta + \frac{1}{2}T\Delta^2 = 0$, the quadratic formula would give $\Delta = \frac{-H \pm \sqrt{H^2 - 2TS}}{T}$. One of these roots is the root near zero that corresponds to the MLE near θ_0 . But this expression is inconvenient for asymptotic analysis. Instead, we solve perturbatively. We already know the first-order answer: $\Delta_1 = -S/H$. Now ask: how much does curvature move this approximate root? Write $\Delta = \Delta_1 + \Delta_2$, where $\Delta_1 = O_p(n^{-1/2})$ and we expect the correction Δ_2 to be smaller.

1.8.2.6 Substitute the first- and second-order pieces

Start with $0 = S + H\Delta + \frac{1}{2}T\Delta^2$. Substitute $\Delta = \Delta_1 + \Delta_2$: $0 = S + H(\Delta_1 + \Delta_2) + \frac{1}{2}T(\Delta_1 + \Delta_2)^2$. Expand the square:

$$0 = S + H\Delta_1 + H\Delta_2 + \frac{1}{2}T(\Delta_1^2 + 2\Delta_1\Delta_2 + \Delta_2^2).$$

But Δ_1 was specifically chosen to solve the linear equation $S + H\Delta_1 = 0$. Therefore those terms disappear: $0 \approx H\Delta_2 + \frac{1}{2}T\Delta_1^2 + T\Delta_1\Delta_2 + \frac{1}{2}T\Delta_2^2$.

1.8.2.7 Keep only terms needed for second order

We know $\Delta_1 = O_p(n^{-1/2})$. We will find that $\Delta_2 = O_p(n^{-1})$. Therefore, $\Delta_1^2 = O_p(n^{-1})$, while $\Delta_1\Delta_2 = O_p(n^{-3/2})$ and $\Delta_2^2 = O_p(n^{-2})$. So, at the relevant order, ignore the latter

two contributions: $0 \approx H\Delta_2 + \frac{1}{2}T\Delta_1^2$. Hence $\Delta_2 \approx -\frac{T}{2H}\Delta_1^2$. Now substitute $\Delta_1 = -S/H$.

Then $\Delta_1^2 = S^2/H^2$. Therefore, $\Delta_2 \approx -\frac{TS^2}{2H^3}$. And this is exactly where the $S_n(\theta_0)^2$ comes from. It appears because the quadratic curvature term acts on the square of the first-order displacement.

1.8.2.8 Geometric interpretation of the squared score

This is perhaps the most important point. The first-order displacement is $\Delta_1 = -S/H$. So the random score S tells us how far the tangent-line root moves from θ_0 . Now curvature evaluates $\frac{1}{2}T\Delta_1^2$. But because $\Delta_1 \propto S$, we necessarily get $\Delta_1^2 \propto S^2$. Thus

$$\text{score fluctuation } S \longrightarrow \text{first-order root displacement } \Delta_1 \propto S$$

and then $\text{curvature acting on displacement } T\Delta_1^2 \propto TS^2$. That is the origin of the squared score.

1.8.2.9 Why does this create bias?

Now the difference between first and second order becomes visually intuitive. Suppose the score at θ_0 fluctuates randomly: $S > 0$ in some samples and $S < 0$ in others. Because $E[S] = 0$, the first-order movements approximately cancel: $E[\Delta_1] \approx 0$. But curvature depends on S^2 . Whether $S = +5$ or $S = -5$, we have $S^2 = 25$. Therefore the curvature correction does not necessarily reverse direction when the score changes sign. That is the geometric origin of bias. For a curved score function, an equally sized positive and negative vertical displacement need not produce equally sized positive and negative horizontal root movements.

1.8.2.10 A simpler geometric analogy

Imagine solving $g(\theta) = 0$. Suppose the graph of g were perfectly linear near the root. Then a vertical perturbation of $+a$ moves the root horizontally one way by some distance d , while a perturbation of $-a$ moves it the other way by exactly d . They cancel on average. But if g is curved, the geometry is asymmetric: $+a \Rightarrow +d_1$, whereas $-a \Rightarrow -d_2$, with generally $d_1 \neq d_2$. So even though the vertical disturbances average to zero, $E[S] = 0$, the horizontal root movements need not: $E[\hat{\theta} - \theta_0] \neq 0$. That is MLE bias generated by likelihood curvature.

1.8.2.11 An additional complication: the random Hessian

The expression $\Delta_2 \approx -TS^2/(2H^3)$ captures one major source of second-order bias, but it is not the complete second-order MLE expansion. Why? Because the Hessian itself is random: $H_n = E[H_n] + \{H_n - E[H_n]\}$. We replaced it informally by approximately $-nI(\theta_0)$, but its random fluctuation also interacts with the score. Consequently, the full second-order expansion contains terms involving things such as S^2 , $S\{H - E[H]\}$, and derivatives of the likelihood. Those interactions also can have nonzero expectations. But the simplified TS^2 derivation is excellent for understanding why second-order terms generate bias at all.

1.8.2.12 Connection to the delta method

This is essentially the same phenomenon discussed earlier for the delta method: $g(\hat{\theta}) \approx g(\theta_0) + g'(\theta_0)\Delta + \frac{1}{2}g''(\theta_0)\Delta^2$. The first-order term is $g'(\theta_0)\Delta$. If $E[\Delta] \approx 0$, its expectation vanishes. But $E[\Delta^2] > 0$. Therefore curvature gives $\frac{1}{2}g''(\theta_0)E[\Delta^2]$, which produces bias. For the MLE score equation, we are doing something closely related but slightly different: $S_n(\theta_0 + \Delta) \approx S + H\Delta + \frac{1}{2}T\Delta^2$. Here we are not merely evaluating a nonlinear function; we are finding the zero of a nonlinear function. First, $S \rightarrow \Delta_1 = -H^{-1}S$. Then curvature acts on this displacement: $T\Delta_1^2$. Since $\Delta_1^2 \propto S^2$, we obtain the second-order correction. The geometry can be summarized as

random score at θ_0	$\rightarrow$	tangent-line root	$\rightarrow$	curvature correction
S		$\Delta_1 \propto S$		$\Delta_2 \propto S^2$

The first-order approximation asks, “Where would the tangent line cross zero?” The second-order approximation asks, “How much does the curvature of the actual score curve move that tangent-line zero?” Because that curvature correction depends on the square of the first-order movement, symmetric score fluctuations can translate into asymmetric root locations—and hence finite-sample MLE bias.

1.8.3 What do the score and Hessian terms mean in a single sample?

Motivating question: what does $S_n(\theta_0)$ correspond to concretely, in one realized sample, and why does it produce a linear shift in $\hat{\theta}$? The key is to give $S_n(\theta_0)$ a very concrete meaning within one realized sample. Suppose we collect one sample $Y_1, \dots, Y_n$. That sample produces its own log-likelihood curve $\ell_n(\theta) = \sum_{i=1}^n \log f(Y_i; \theta)$ and its own score curve $S_n(\theta) = \ell'_n(\theta)$. The MLE $\hat{\theta}$ is where this sample-specific score equals zero: $S_n(\hat{\theta}) = 0$.

1.8.3.1 What exactly is $S_n(\theta_0)$?

Take the true population parameter θ_0 , but evaluate the sample’s score there:

$$S_n(\theta_0) = \left. \frac{\partial \ell_n(\theta)}{\partial \theta} \right|_{\theta=\theta_0}.$$

It answers a very intuitive question: for this particular random sample, what is the slope of the log-likelihood when we stand at the true parameter θ_0 ? If $S_n(\theta_0) = 0$, then this particular sample happens to have its likelihood locally maximized at θ_0 , so approximately $\hat{\theta} = \theta_0$. But typically, because of sampling randomness, $S_n(\theta_0) \neq 0$. If $S_n(\theta_0) > 0$, the sample log-likelihood is still increasing at θ_0 , so its maximum tends to lie to the right: $\hat{\theta} > \theta_0$. If $S_n(\theta_0) < 0$, the likelihood is decreasing at θ_0 , so its maximum tends to lie to the left: $\hat{\theta} < \theta_0$. So $S_n(\theta_0)$ is the sample’s random push away from the true parameter.

1.8.3.2 A concrete example: the normal mean

Take $Y_i \sim N(\mu, \sigma^2)$, with σ^2 known. The true parameter is μ_0 . The score is $S_n(\mu) = \frac{1}{\sigma^2} \sum_{i=1}^n (Y_i - \mu)$. Evaluate it at the true mean: $S_n(\mu_0) = \frac{1}{\sigma^2} \sum_{i=1}^n (Y_i - \mu_0)$. Equivalently,

$S_n(\mu_0) = \frac{n}{\sigma^2}(\bar{Y} - \mu_0)$. Now its meaning is transparent. Suppose the true mean is $\mu_0 = 100$. One random sample might happen to have $\bar{Y} = 102$. Then $S_n(\mu_0) > 0$: at $\mu = 100$, increasing μ would improve the likelihood, so the MLE moves upward, $\hat{\mu} = 102$. Another sample might give $\bar{Y} = 98$; then $S_n(\mu_0) < 0$, and the likelihood pushes the estimate downward, $\hat{\mu} = 98$. Across repeated samples, $E[S_n(\mu_0)] = 0$, so there is no systematic first-order push left or right.

1.8.3.3 What does $H_n(\theta_0)(\hat{\theta} - \theta_0)$ mean?

Let $H_n(\theta_0) = S'_n(\theta_0) = \ell''_n(\theta_0)$. The Hessian tells us how quickly the score changes when we move θ . For a scalar parameter, $H_n(\theta_0)$ is the change in score per change in parameter, locally. Therefore, if we move from θ_0 to $\hat{\theta} = \theta_0 + \Delta$, the approximate change in the score is $H_n(\theta_0)\Delta$. So the Taylor approximation $S_n(\hat{\theta}) \approx S_n(\theta_0) + H_n(\theta_0)(\hat{\theta} - \theta_0)$ has a very simple interpretation: score at new point $\approx$ score at starting point + change in score caused by moving. But at the MLE, $S_n(\hat{\theta}) = 0$. Therefore, $0 \approx S_n(\theta_0) + H_n(\theta_0)(\hat{\theta} - \theta_0)$. This says: start with the nonzero score created by sampling randomness at θ_0 , then move θ until the resulting change in the score exactly cancels that initial score.

1.8.3.4 Return to the normal example

For the normal mean, $S_n(\mu) = \frac{n}{\sigma^2}(\bar{Y} - \mu)$. Therefore, $H_n(\mu) = -n/\sigma^2$. Suppose again $\mu_0 = 100$, $\bar{Y} = 102$. Then $S_n(100) = 2n/\sigma^2$. We need to move from 100 to 102: $\Delta = \hat{\mu} - \mu_0 = 2$. That movement changes the score by $H_n\Delta = -\frac{n}{\sigma^2}(2) = -2n/\sigma^2$. Therefore,

$$\underbrace{\frac{2n}{\sigma^2}}_{\text{initial score at 100}} + \underbrace{\left(-\frac{2n}{\sigma^2}\right)}_{\text{score change from moving to 102}} = 0.$$

Exactly. So $S_n(\theta_0)$ is the initial random score imbalance created by the sample, while $H_n(\theta_0)(\hat{\theta} - \theta_0)$ is the approximate change in score produced by moving the parameter from the truth toward the sample MLE.

1.8.3.5 A force-and-stiffness analogy

Think of $S_n(\theta_0)$ as a random force exerted by the particular sample. The Hessian measures the stiffness of the likelihood around θ_0 . Then $\hat{\theta} - \theta_0$ is the displacement produced by that force. The first-order score equation is $0 \approx S_n(\theta_0) + H_n(\theta_0)(\hat{\theta} - \theta_0)$. Hence $\hat{\theta} - \theta_0 \approx -H_n(\theta_0)^{-1}S_n(\theta_0)$. So MLE displacement = random sample push / likelihood curvature (stiffness), roughly speaking. A very sharply curved likelihood has large $|H_n|$, so the same random score produces only a small movement in $\hat{\theta}$. A flat likelihood has small $|H_n|$, so the same random score produces a much larger movement. This is precisely where Fisher information enters: more Fisher information means greater expected likelihood curvature, hence less movement of the MLE in response to sample randomness.

1.8.3.6 Now the second-order term becomes intuitive

The linear story assumes the stiffness is essentially constant as we move from θ_0 toward $\hat{\theta}$. But generally the score curve itself is curved, so $H_n(\theta)$ changes as θ changes. Hence we improve $0 \approx S_n(\theta_0) + H_n(\theta_0)\Delta$ to $0 \approx S_n(\theta_0) + H_n(\theta_0)\Delta + \frac{1}{2}T_n(\theta_0)\Delta^2$, where $T_n(\theta_0) = S_n''(\theta_0)$. Now the three pieces have a very concrete interpretation:

$S_n(\theta_0) = \text{random score imbalance created by this sample,}$ $H_n(\theta_0)\Delta = \text{linear change in score caused by moving } \theta,$ $\frac{1}{2}T_n(\theta_0)\Delta^2 = \text{correction because the score's slope changes while we move.}$

And this last term is where the asymmetry—and therefore the $O(1/n)$ bias—can arise. One small refinement: sampling randomness creates a nonzero score at the fixed true value θ_0 ; the MLE then shifts away from θ_0 until the score reaches zero, rather than the score "creating a linear shift at $\hat{\theta}$ " directly. That distinction makes the whole Taylor argument much easier to visualize.

1.8.4 Why the normal-mean score is linear but the normal-variance score is not

Motivating question: for the normal mean, the log-likelihood is quadratic in μ , so the score is linear and there is no bias. Can the same be shown for σ , where the score has genuine curvature and bias appears? The normal model is an excellent example for seeing the geometry, but there is one important refinement:

nonlinearity of the score makes second-order bias possible, but does not by itself guarantee bias.
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Let us isolate σ first by assuming μ is known. This makes the comparison with the μ -case very clean. For $Y_i \sim N(\mu, \sigma^2)$, with μ known, define $Q = \sum_{i=1}^n (Y_i - \mu)^2$. The log-likelihood, ignoring constants, is $\ell_n(\sigma) = -n \log \sigma - \frac{Q}{2\sigma^2}$. Compare this with the likelihood as a function of μ : $\ell_n(\mu) = -\frac{1}{2\sigma^2} \sum_i (Y_i - \mu)^2 + C$, which is exactly quadratic in μ . Therefore its score is exactly linear in μ . But in σ , $\ell_n(\sigma) = -n \log \sigma - \frac{Q}{2\sigma^2}$, which is clearly nonlinear: it contains both $\log \sigma$ and $1/\sigma^2$.

1.8.4.1 The score for σ

Differentiate: $S_n(\sigma) = \ell'_n(\sigma) = -\frac{n}{\sigma} + \frac{Q}{\sigma^3}$. So $S_n(\sigma) = -\frac{n}{\sigma} + \frac{Q}{\sigma^3}$. This score is not linear in σ . The MLE solves $S_n(\hat{\sigma}) = 0$, so $-\frac{n}{\hat{\sigma}} + \frac{Q}{\hat{\sigma}^3} = 0$. Multiply by $\hat{\sigma}^3$: $-n\hat{\sigma}^2 + Q = 0$.

Therefore $\hat{\sigma}^2 = \frac{Q}{n}$, $\hat{\sigma} = \sqrt{\frac{Q}{n}}$. Already you can see the source of nonlinearity: $Q \rightarrow \hat{\sigma} = \sqrt{Q/n}$.

1.8.4.2 What is $S_n(\sigma_0)$ in one sample?

Let the true standard deviation be σ_0 . Then $S_n(\sigma_0) = -\frac{n}{\sigma_0} + \frac{Q}{\sigma_0^3}$. Rewrite:

$S_n(\sigma_0) = \frac{Q - n\sigma_0^2}{\sigma_0^3}$. This is very interpretable. Under the true model, $E[Q] = n\sigma_0^2$ when μ is known. Therefore $S_n(\sigma_0)$ measures whether the sample happened to have more or less squared dispersion than expected. If $Q > n\sigma_0^2$, then $S_n(\sigma_0) > 0$: at the true σ_0 , the likelihood would improve if σ increased, so typically $\hat{\sigma} > \sigma_0$. If $Q < n\sigma_0^2$, then $S_n(\sigma_0) < 0$, and the sample pushes the MLE downward. So this is directly analogous to the normal-mean case. For the mean, $S_n(\mu_0) = \frac{n}{\sigma^2}(\bar{Y} - \mu_0)$, so the random push comes from the sample mean. For standard deviation, $S_n(\sigma_0) = \frac{Q - n\sigma_0^2}{\sigma_0^3}$, so the random push comes from the sample's squared dispersion.

1.8.4.3 Taylor-expanding the score around σ_0

Write $\Delta = \hat{\sigma} - \sigma_0$. Since $S_n(\hat{\sigma}) = 0$, expand around σ_0 : $0 = S_n(\sigma_0) + S'_n(\sigma_0)\Delta + \frac{1}{2}S''_n(\sigma_0)\Delta^2 + \dots$. We have $S_n(\sigma) = -n/\sigma + Q/\sigma^3$. Therefore, $S'_n(\sigma) = \frac{n}{\sigma^2} - \frac{3Q}{\sigma^4}$. And $S''_n(\sigma) = -\frac{2n}{\sigma^3} + \frac{12Q}{\sigma^5}$. Thus the second-order score equation is

$$0 \approx \left(-\frac{n}{\sigma_0} + \frac{Q}{\sigma_0^3}\right) + \left(\frac{n}{\sigma_0^2} - \frac{3Q}{\sigma_0^4}\right)\Delta + \frac{1}{2}\left(-\frac{2n}{\sigma_0^3} + \frac{12Q}{\sigma_0^5}\right)\Delta^2.$$

This is exactly the quadratic approximation of interest.

1.8.4.4 Compare this directly with the μ -case

For μ , with σ known, $S_n(\mu) = \frac{n}{\sigma^2}(\bar{Y} - \mu)$. Therefore $S'_n(\mu) = -n/\sigma^2$ and $S''_n(\mu) = 0$. So the Taylor expansion ends exactly after the linear term: $0 = S_n(\mu_0) + S'_n(\mu_0)(\hat{\mu} - \mu_0)$. There is literally no curvature correction. Hence $\hat{\mu} - \mu_0 = -\frac{S_n(\mu_0)}{S'_n(\mu_0)} = \bar{Y} - \mu_0$. And because $E[S_n(\mu_0)] = 0$, we get exact unbiasedness. For σ , $S''_n(\sigma_0) \neq 0$. So the score curve itself is curved, and the tangent-line root is not the exact root. That is where a second-order correction becomes relevant.

1.8.4.5 First-order solution for $\hat{\sigma} - \sigma_0$

Ignore the quadratic term first: $0 \approx S_0 + H_0\Delta_1$, where $S_0 = S_n(\sigma_0)$, $H_0 = S'_n(\sigma_0)$. Then $\Delta_1 = -S_0/H_0$. For asymptotic analysis, replace the Hessian by its expectation. Since $E[Q] = n\sigma_0^2$, $E[H_0] = \frac{n}{\sigma_0^2} - \frac{3n\sigma_0^2}{\sigma_0^4} = -\frac{2n}{\sigma_0^2}$. This is negative, as expected near the likelihood

maximum. Thus approximately, $\Delta_1 \approx \frac{\sigma_0^2}{2n}S_0$. Since $S_0 = \frac{Q - n\sigma_0^2}{\sigma_0^3}$, we get $\Delta_1 \approx \frac{Q - n\sigma_0^2}{2n\sigma_0}$.

This is the first-order random movement in the MLE. Its expectation is zero because $E[Q - n\sigma_0^2] = 0$. Therefore, $E[\Delta_1] = 0$. Exactly the same story as before: the first-order score fluctuation is centered.

1.8.4.6 Second-order correction

Now write $\Delta = \Delta_1 + \Delta_2$. The approximate score equation is $0 \approx S_0 + H_0\Delta + \frac{1}{2}T_0\Delta^2$. Because $S_0 + H_0\Delta_1 = 0$, the next-order terms give $H_0\Delta_2 + \frac{1}{2}T_0\Delta_1^2 \approx 0$. Therefore,

$\Delta_2 \approx -\frac{T_0}{2H_0}\Delta_1^2$. Or, $\Delta_2 \approx -\frac{T_0 S_0^2}{2H_0^3}$. Now the critical difference appears: $E[S_0] = 0$, but $E[S_0^2] > 0$. So the second-order displacement does not generally average to zero. That is the mechanism discussed earlier.

1.8.4.7 A more direct route via Jensen's inequality

There is a simpler route because we know the exact MLE: $\hat{\sigma} = \sqrt{Q/n}$. Define $V = Q/n$. When μ is known, $E[V] = \sigma_0^2$. Thus V itself is unbiased for σ_0^2 . But $\hat{\sigma} = \sqrt{V}$. The square-root function is concave: $g(v) = \sqrt{v}$, $g''(v) = -\frac{1}{4v^{3/2}} < 0$. Therefore Jensen gives $E[\sqrt{V}] < \sqrt{E[V]} = \sigma_0$. Hence $E[\hat{\sigma}] < \sigma_0$. So the MLE of σ is downward biased. This is the same curvature effect viewed from a different direction.

1.8.4.8 Computing the approximate bias

For known μ , $V = \frac{1}{n} \sum_i (Y_i - \mu)^2$. We know $E[V] = \sigma_0^2$ and, because a squared normal has variance $2\sigma_0^4$, $\text{Var}(V) = \frac{2\sigma_0^4}{n}$. Apply the second-order delta method to $\hat{\sigma} = g(V) = \sqrt{V}$. We have $g'(\sigma_0^2) = \frac{1}{2\sigma_0}$, and $g''(\sigma_0^2) = -\frac{1}{4\sigma_0^3}$. Since V is unbiased, $\text{Bias}(\hat{\sigma}) \approx \frac{1}{2}g''(\sigma_0^2)\text{Var}(V)$. Thus

$$\begin{aligned}\text{Bias}(\hat{\sigma}) &\approx \frac{1}{2} \left(-\frac{1}{4\sigma_0^3} \right) \frac{2\sigma_0^4}{n} \\ &= -\frac{\sigma_0}{4n}.\end{aligned}$$

Therefore,

$$\text{Bias}(\hat{\sigma}) \approx -\frac{\sigma_0}{4n} \quad (\mu \text{ known}).$$

This is an $O(1/n)$ downward bias.

1.8.4.9 A nuance about parameterization

This is actually a very useful lesson. If we parameterize the normal model using $v = \sigma^2$, then, with μ known, $\hat{v} = \frac{1}{n} \sum_i (Y_i - \mu)^2$ is exactly unbiased: $E[\hat{v}] = v$. Yet $\hat{\sigma} = \sqrt{\hat{v}}$ is biased. So, nonlinear likelihood/score $\nRightarrow$ MLE necessarily biased. The bias also depends on which parameterization we estimate. For known μ , $\hat{\sigma}^2$ is unbiased, while $\hat{\sigma}$ is biased. This is because unbiasedness is not invariant to nonlinear transformations.

1.8.4.10 When μ is also unknown

Now we get the familiar additional bias. The joint MLEs are $\hat{\mu} = \bar{Y}$ and $\hat{\sigma}^2 = \frac{1}{n} \sum_i (Y_i - \bar{Y})^2$.

Then $E[\hat{\sigma}^2] = \frac{n-1}{n}\sigma_0^2$. So even the variance MLE is downward biased: $\text{Bias}(\hat{\sigma}^2) = -\frac{\sigma_0^2}{n}$.

This particular bias has an additional source: estimating μ consumes one degree of freedom. Then taking $\hat{\sigma} = \sqrt{\hat{\sigma}^2}$ adds another nonlinear-transformation effect. So with both μ and σ unknown, there are really two effects: estimating μ already makes $\hat{\sigma}^2$ downward biased, and then $\hat{\sigma} = \sqrt{\hat{\sigma}^2}$ adds an additional concavity bias. This normal example is therefore especially instructive: the μ -likelihood is quadratic, so the score equation is exactly linear and the root responds linearly to sample noise; the σ -likelihood has curvature beyond

quadratic, so the score root responds nonlinearly, and the second-order score expansion captures the resulting asymmetric displacement.

1.9 Estimation error, bias, and mean squared error

1.9.1 Estimators vary across samples

An estimator is a random variable because a new random sample would generally produce a different estimate. For one realized sample, the estimation error is $\hat{\theta}$ minus the unknown true value θ_0 . The lecture illustrates four possibilities by combining high or low variability with biased or unbiased centering.

$$\text{Estimation error} = \hat{\theta} - \theta_0, \quad \text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta_0$$

Equation 14. Error and bias

$$\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta_0)^2] = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2$$

Equation 15. Bias-variance decomposition

Unbiasedness is desirable but not sufficient: two unbiased estimators may have very different variances. MSE supplies a single criterion that balances systematic error and sampling variability.

1.9.2 Consistency and the root-n rate

Consistency means that the estimator converges in probability to the true parameter as sample size grows. It is stronger and more precise than saying only that a displayed confidence interval becomes narrower.

$$\hat{\theta}_n \xrightarrow{P} \theta_0 \quad \text{means} \quad P(|\hat{\theta}_n - \theta_0| > \varepsilon) \rightarrow 0$$

Equation 16. Definition of consistency

Qualification to the lecture shorthand. MLEs are consistent and asymptotically efficient under standard regularity conditions and correct model specification. They are not universally consistent for every likelihood, boundary problem, unidentified model, mixture model, or misspecified data-generating process.

1.10 Fisher information, curvature, and standard errors

1.10.1 Why a sharper likelihood means greater precision

Near its maximum, a sharply curved log-likelihood changes quickly when θ moves away from $\hat{\theta}$. The data therefore discriminate strongly among nearby parameter values. A

flatter log-likelihood contains less local information and produces a larger standard error.

$$j_n(\theta) = -\frac{\partial^2 \mathcal{L}(\theta)}{\partial \theta^2}, \quad I_n(\theta) = E[j_n(\theta)]$$

Equation 17. Observed and expected Fisher information

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, I_1^{-1}(\theta_0))$$

Equation 18. Asymptotic normality at the root-n rate

$$\text{Var}(\hat{\theta}) \approx I_n^{-1}(\hat{\theta}), \quad SE(\hat{\theta}_j) \approx \sqrt{[I_n^{-1}]_{jj}}$$

Equation 19. Information-based covariance and standard error

Applied to the McGill normal fit above, the observed information can be read directly off the optimizer's Hessian, reproducing the standard error that `summary()` already printed:

```
import math

I_mean = -result_mcgill.model.hessian(result_mcgill.params)[0, 0]
SE_mean = math.sqrt(1 / I_mean)
MLE_mean = result_mcgill.params[0]

from scipy.stats import norm
CV_LCL, CV_UCL = norm.ppf(0.025), norm.ppf(0.975)
Interval_LCL = MLE_mean + CV_LCL * SE_mean
Interval_UCL = MLE_mean + CV_UCL * SE_mean
```

This gives $I_n(\hat{\mu}) \approx 3.851 \times 10^{-8}$, $SE(\hat{\mu}) = \sqrt{1/I_n(\hat{\mu})} \approx \$5,095.8$, and a hand-built 95% Wald interval of [\$99,040.7, \$119,015.9] — matching, to rounding, both the std err column and the [0.025, 0.975] column that `result_mcgill.summary()` produced automatically. Equation 19 is therefore not just notation: it is exactly what the summary table computes internally.

1.10.2 Bernoulli standard error and confidence interval

$$I_n(p) = \frac{n}{p(1-p)}, \quad SE(\hat{p}) \approx \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Equation 20. Bernoulli information and plug-in standard error

$$\hat{p} \pm z_{1-\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Equation 21. Large-sample Wald interval

Table 4. Precision improvement when the observed proportion stays at 0.60

Sample size n	Estimated SE	Approximate 95% interval	Width
10	0.1549	[0.296, 0.904]	0.607
100	0.0490	[0.504, 0.696]	0.192
1,000	0.0155	[0.570, 0.630]	0.061

Increasing n from 10 to 100 multiplies the sample size by 10 but reduces the standard error only by $\sqrt{10}$, about 3.16. This is the root- n rate emphasized in the lecture. The Wald interval is pedagogically useful here but can perform poorly near $p = 0$ or $p = 1$ or with very small samples; score or exact intervals may be preferable.

1.10.3 Normal-model information

For independent normal observations, the mean and scale have orthogonal information under the usual parameterization. The leading standard-error approximations are:

$$SE(\hat{\mu}) \approx \frac{\sigma}{\sqrt{n}}, \quad SE(\hat{\sigma}) \approx \frac{\sigma}{\sqrt{2n}}$$

Equation 22. Normal-model standard errors

1.11 Pooling McGill and Toronto salary samples

1.11.1 Common-distribution model

The lecture proposes adding the McGill and Toronto log-likelihoods. This is mathematically equivalent to pooling the two samples only if both groups are assumed to share the same mean and standard deviation. Under independence, log-likelihood contributions add.

$$\mathcal{L}_{total}(\mu, \sigma) = \mathcal{L}_{McGill}(\mu, \sigma) + \mathcal{L}_{Toronto}(\mu, \sigma)$$

Equation 23. Pooled log-likelihood under common parameters

The accompanying notebook implements exactly this sum, extending the single-sample class to accept two datasets and adding their log-likelihood contributions under one shared (mean, std):

```
class MLE(GenericLikelihoodModel):
    def __init__(self, data1, data2):
        self.data1 = data1
        self.data2 = data2
        super().__init__(endog=np.zeros(1))
        self.nparams = 2
        self._set_extra_params_names(["mean", "std"])
        self._eps = 1e-9
    def loglike(self, params):
```

```

mean, std = params
if not (self._eps < std):
    return -np.inf
l11 = norm.logpdf(self.data1, loc=mean, scale=std).sum()
l12 = norm.logpdf(self.data2, loc=mean, scale=std).sum()
return l11 + l12

mcgill = df["McGill"].to_numpy()
toronto = df["Toronto"].to_numpy()
result_pooled = MLE(mcgill, toronto).fit(start_params=[0, 10],
method="bfgs", maxiter=200)
print(result_pooled.summary())

```

coef	std err	z	P> z	[0.025	0.975]
mean	1.157e+05	3545.387	32.641	0.000	1.09e+05 1.23e+05
std	5.014e+04	2508.301	19.989	0.000	4.52e+04 5.51e+04

Log-Likelihood: -2448.2 AIC: 4900. BIC: 4896.

The pooled mean, about \$115,700, sits between McGill's own MLE mean of \$109,028 and Toronto's raw sample mean of \$122,385, and the pooled standard error on the mean (\$3,545) is noticeably tighter than McGill's standalone standard error (\$5,096) — exactly the precision gain the pooling principle promises, provided the shared-mean assumption is actually defensible for these two schools.

1.11.2 Separate means with a common standard deviation

A more flexible model allows each school to have its own mean while retaining a shared standard deviation. For McGill observations x_i and Toronto observations y_j :

$$x_i \stackrel{\text{ind}}{\sim} N(\mu_M, \sigma^2), \quad y_j \stackrel{\text{ind}}{\sim} N(\mu_T, \sigma^2)$$

Equation 24. Two means and one common variance

$$\hat{\sigma}^2 = \frac{\sum_i^{n_M} (x_i - \bar{x})^2 + \sum_j^{n_T} (y_j - \bar{y})^2}{n_M + n_T}$$

Equation 25. MLE of the common variance

Table 5. Modeling choices for multiple schools

Model	Parameters	Advantage	Primary risk
Complete pooling	One common μ and σ	Highest nominal precision	Bias if school means differ
Separate means, common scale	μ_M, μ_T, σ	Allows center differences	Shared dispersion may be false
Separate distributions	$\mu_M, \sigma_M, \mu_T, \sigma_T$	Most flexible of these three	More parameters and larger uncertainty

Pooling principle. More observations reduce variance only when the observations are informative about the same target parameter. Pooling heterogeneous populations can trade lower variance for substantial bias.

1.12 Reading a maximum-likelihood output table

Table 6. Interpretation guide for the summary output

Field	Meaning	Recommended interpretation
coef	Maximum likelihood estimate	Best-fitting parameter value under the stated model
std err	Estimated sampling standard deviation	Typical uncertainty in repeated samples; not the raw-data standard deviation
z	Estimate divided by its standard error	Distance from a null value of zero in standard-error units
P> z	Large-sample two-sided p-value	Evidence against a zero null under the model and regularity assumptions
[0.025, 0.975]	Approximate 95% confidence interval	Range produced by the estimator's repeated-sampling procedure
Log-Likelihood	Maximized log-likelihood	Useful for comparing appropriately related models; larger is better before penalties
Converged / iterations	Optimizer diagnostic	Necessary computational evidence, not proof that the model is scientifically appropriate

1.13 Common errors and diagnostic checks

1. Counting error: verify that the Bernoulli exponents equal the observed numbers of ones and zeros.
 2. Probability/likelihood confusion: data vary in a probability model; parameters vary in a likelihood function.
 3. Wrong distribution: `bernoulli.logpmf` is for 0/1 outcomes; `norm.logpdf` is for continuous normal modeling.
 4. Wrong scale domain: enforce $\sigma > 0$, preferably by estimating $\log(\sigma)$.
 5. Mean/standard-error confusion: a standard error describes estimator uncertainty, not the dispersion of individual observations.
 6. Blind trust in convergence: test several starting points, inspect gradients, and compare with closed-form results when possible.
 7. Unjustified pooling: test or defend equality restrictions before combining groups under common parameters.
 8. Model misspecification: inspect distributions and use robust uncertainty methods when the likelihood is only an approximation.
-

1.14 Worked practice problems

1.14.1 Another Bernoulli sample

Suppose 14 of 20 independent product tests pass. Find the Bernoulli MLE and its plug-in standard error.

$$\hat{p} = \frac{14}{20} = 0.70, \quad SE \approx \sqrt{\frac{0.70(0.30)}{20}} \approx 0.1025$$

Equation 26. Solution to Problem 1

1.14.2 Normal scale denominator

For observations 2, 4, and 6, compute the normal MLEs of μ and σ squared.

$$\hat{\mu} = 4, \quad \hat{\sigma}^2 = \frac{(2-4)^2 + (4-4)^2 + (6-4)^2}{3} = \frac{8}{3}$$

Equation 27. Solution to Problem 2

1.14.3 Sample-size planning by relative precision

How must sample size change to cut a root-n standard error in half?

$$\frac{c}{\sqrt{n_{new}}} = \frac{c}{2\sqrt{n_{old}}} \Rightarrow n_{new} = 4n_{old}$$

Equation 28. Solution to Problem 3

1.15 Compact mathematical reference

Table 7. Core formulas from the lecture and its extensions

Concept	Formula	Use
Bernoulli likelihood	$p^S(1-p)^{n-S}$	Estimate a binary-event probability
Bernoulli MLE	$\hat{p} = S/n$	Sample proportion
Normal MLEs	$\hat{\mu} = \bar{x}; \hat{\sigma}^2 = \text{SSE}/n$	Estimate normal location and scale
Bias	$E[\hat{\theta}] - \theta$	Systematic estimation error
MSE	Variance + Bias ²	Overall squared-error criterion
Observed information	$-\mathcal{L}''(\theta)$	Local log-likelihood curvature
Asymptotic covariance	$I_n(\hat{\theta})^{-1}$	Approximate uncertainty matrix
Root-n rate	SE proportional to $n^{-1/2}$	Precision improves with diminishing returns

1.16 Case studies: The Maximum Likelihood Principle

Let $y_1, \dots, y_n$ be independent and identically distributed (i.i.d.) observations drawn from a density (or probability mass function) $f(y \mid \theta)$, indexed by an unknown parameter vector θ . The **likelihood function** is the joint density of the data viewed as a function of θ ,

$$L(\theta \mid y_1, \dots, y_n) = \prod_{i=1}^n f(y_i \mid \theta),$$

and the **maximum likelihood estimator** (MLE) is the value $\hat{\theta}$ that maximizes $L(\theta)$, or equivalently, the **log-likelihood**

$$\ell(\theta) = \sum_{i=1}^n \log f(y_i \mid \theta), \quad \hat{\theta} = \arg \max_{\theta} \ell(\theta).$$

In all four case studies below we do *not* derive $\hat{\theta}$ analytically; instead we hand $\ell(\theta)$ to a numerical optimizer (the BFGS quasi-Newton algorithm) and let it climb the log-likelihood surface. This mirrors how MLE is used in practice whenever the score equations $\partial \ell / \partial \theta = 0$ have no closed-form solution, or simply as a teaching device to make the optimization explicit.

Two generic building blocks are common to every notebook:

- (i) A custom class that inherits from `GenericLikelihoodModel` and overrides `loglike(self, params)` to return $\ell(\theta)$ for a candidate parameter vector `params`.
- (ii) A call to `model.fit(start_params=..., method="bfgs")`, which numerically maximizes $\ell(\theta)$ and returns a `results` object with point estimates, standard errors (from the numerically evaluated Hessian / observed Fisher information), z -statistics, and p -values — formatted exactly as a standard regression summary table.

Because the estimated standard errors come from the curvature (Hessian) of the log-likelihood at $\hat{\theta}$, they rely on the classical large-sample result

$$\hat{\theta} \stackrel{a}{\sim} \mathcal{N}(\theta_0, \mathcal{I}(\hat{\theta})^{-1}), \quad \mathcal{I}(\theta) = -\mathbb{E} \left[\frac{\partial^2 \ell(\theta)}{\partial \theta \partial \theta^\top} \right],$$

where $\mathcal{I}(\theta)$ is the **Fisher information**. In practice $\mathcal{I}(\theta)$ is replaced by the *observed* information $-\partial^2 \ell(\hat{\theta}) / \partial \theta \partial \theta^\top$ (the negative Hessian evaluated at $\hat{\theta}$), which `statsmodels` computes numerically. This is exactly the machinery used to build the confidence interval in Case Study 3.

1.17 Case Study 1: MLE of a Bernoulli Parameter (Coin Flips)

1.17.1 Setting and data

A coin is flipped $n = 100$ times, recorded as $y_i \in \{0, 1\}$ (tail/heads) in the column `Coins` of `M3 Coin flip data.xlsx`. The sample mean is $\bar{y} = 0.60$, so on average the coin lands heads 60% of the time — suggesting the coin may be biased.

1.17.2 Statistical model

Each flip is modeled as an i.i.d. Bernoulli trial, $Y_i \sim \text{Bernoulli}(p)$, with probability mass function $f(y | p) = p^y(1 - p)^{1-y}$ for $y \in \{0, 1\}$. The log-likelihood for the sample is

$$\ell(p) = \sum_{i=1}^n \left[y_i \log p + (1 - y_i) \log(1 - p) \right], \quad p \in (0, 1).$$

Setting $\ell'(p) = 0$ gives the familiar closed-form MLE $\hat{p} = \bar{y}$; the notebook instead recovers this numerically, which also automatically returns a standard error via the Hessian.

1.17.3 Python implementation

```

1 import pandas as pd
2 import numpy as np
3 from statsmodels.base.model import GenericLikelihoodModel
4 from scipy.stats import bernoulli
5
6 # --- Load data ---
7 df = pd.read_excel("M3 Coin Dataset.xlsx", sheet_name="Sheet1")
8 data = df["Coins"].to_numpy() # array of 0/1 outcomes, n = 100
9
10 # --- Custom likelihood model ---
11 class MLE(GenericLikelihoodModel):
12     def __init__(self, data):
13         self.y = data
14         super().__init__(endog=np.zeros(1))
15         self.nparams = 1
16         self._set_extra_params_names(["p"])
17         self._eps = 1e-9
18
19     def loglike(self, params):
20         p = params
21         if not (self._eps < p < 1 - self._eps): # keep p inside (0,1)
22             return -np.inf
23         return bernoulli.logpmf(self.y, p).sum()
24
25 # --- Fit by numerical maximization (BFGS) ---
26 model = MLE(data)
27 result = model.fit(start_params=[0.5], method="bfgs", maxiter
28                    =200)
29 print(result.summary())

```

Implementation notes

- `loglike` must return $-\infty$ (not raise an error) outside the admissible range $p \in (0, 1)$, so the optimizer is repelled from invalid regions rather than crashing.
- `bernoulli.logpmf(self.y, p).sum()` vectorizes $\sum_i \log f(y_i | p)$ in one line using `scipy.stats`.
- `start_params=[0.5]` is the optimizer's initial guess; BFGS uses local gradient information to climb to the maximum.

1.17.4 Output

```

1 Optimization terminated successfully.
2 Current function value: 67.301167
3 Iterations: 4
4 Function evaluations: 8
5 Gradient evaluations: 6
6
7 coef      stderr   z           P>|z|     [0.025      0.975]
8 -----
9 p      0.6000   0.049    12.247     0.000     0.504      0.696

```

1.17.5 Interpretation

The MLE is $\hat{p} = 0.600$, exactly the sample proportion of heads, with an estimated standard error of 0.049. The associated z -statistic ($z = 12.247$, $p < 0.001$) tests $H_0 : p = 0$, which is not the interesting null here; a more relevant exercise for students is to test $H_0 : p = 0.5$ using $\hat{p}$ and its standard error, i.e. $z = (0.60 - 0.50)/0.049 \approx 2.04$, which is (marginally) significant at the 5% level — suggestive but not overwhelming evidence that the coin is unfair.

1.18 Case Study 2: MLE of a Normal Distribution (Vehicle Fuel Efficiency)

1.18.1 Setting and data

The *Auto* data set (M3 `Auto_data.xlsx`, $n = 392$ vehicles) records fuel efficiency in miles per gallon (`mpg`) along with engine characteristics. We treat `mpg` as a continuous random variable and estimate the two parameters of a Normal distribution that best explains its dispersion.

1.18.2 Statistical model

Assume $Y_i \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with density

$$f(y \mid \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(y - \mu)^2}{2\sigma^2}\right).$$

The log-likelihood is

$$\ell(\mu, \sigma) = \sum_{i=1}^n \log f(y_i \mid \mu, \sigma) = -\frac{n}{2} \log(2\pi) - n \log \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2.$$

The classical closed-form maximizers are $\hat{\mu} = \bar{y}$ and $\hat{\sigma} = \sqrt{\frac{1}{n} \sum_i (y_i - \bar{y})^2}$ (the *population*, i.e. n -denominator, standard deviation — not the sample standard deviation with $n - 1$). This is a useful point to flag for students: the MLE of σ is a *biased* estimator of the true population standard deviation in finite samples.

1.18.3 Python implementation

```

1  import pandas as pd
2  import numpy as np
3  from statsmodels.base.model import GenericLikelihoodModel
4  from scipy.stats import norm
5
6  # --- Load data ---
7  df = pd.read_excel("M3 Auto Dataset.xlsx", sheet_name="
      Original Dataset")
8  data = df["mpg"].to_numpy()          # n = 392 observations
9
10 # --- Custom likelihood model with two free parameters ---
11 class MLE(GenericLikelihoodModel):
12     def __init__(self, data):
13         self.y = data
14         super().__init__(endog=np.zeros(1))
15         self.nparams = 2
16         self._set_extra_params_names(["mean", "std"])
17         self._eps = 1e-9
18
19     def loglike(self, params):
20         mean, std = params
21         if not (self._eps < std): # variance must be positive
22             return -np.inf
23         return norm.logpdf(self.y, loc=mean, scale=std).sum()
24
25 # --- Fit ---
26 model = MLE(data)
27 result = model.fit(start_params=[0, 10], method="bfgs", maxiter
      =200)
28 print(result.summary())

```

Implementation notes

- The parameter vector is now two-dimensional, (mean, std); the optimizer searches a 2-D surface instead of a line.
- The guard `self._eps < std` enforces $\sigma > 0$, since `norm.logpdf` is undefined (or $-\infty$) for non-positive scale.
- `start_params=[0, 10]` is a coarse initial guess; BFGS converges to the same optimum regardless, but a reasonable start speeds up convergence.

1.18.4 Output

```

1  Optimization terminated successfully.
2  Current function value: 1361.191347
3  Iterations: 11
4  Function evaluations: 18
5  Gradient evaluations: 15
6
7  coef      std err          z      P>|z|      [0.025      0.975]

```

8	-----						
9	mean	23.4459	0.394	59.551	0.000	22.674	24.218
10	std	7.7950	0.278	28.000	0.000	7.249	8.341

1.18.5 Interpretation

The MLE recovers exactly the sample mean, $\hat{\mu} = 23.4459$, and the population standard deviation, $\hat{\sigma} = 7.7950$ mpg — both match `df.describe()` up to the n vs. $n - 1$ denominator convention. The reported standard errors (e.g. 0.394 for the mean) again come from the negative Hessian of ℓ evaluated at $(\hat{\mu}, \hat{\sigma})$, providing a numerical route to the same asymptotic variance formulas $\text{Var}(\hat{\mu}) \approx \sigma^2/n$ that students may already know from a first course in statistics.

1.19 Case Study 3: Confidence Intervals from the Observed Fisher Information (Graduate Salaries)

1.19.1 Setting and data

M3 Salary Data.xlsx records simulated starting salaries for $n = 100$ graduates from each of three programs (McGill, Toronto, UBC). This case study uses only the McGill column and goes one step further than Case Study 2: after fitting a Normal MLE, it manually reconstructs a 95% confidence interval for μ from *first principles*, using the Hessian of the log-likelihood rather than reading it off `result.summary()`. This is a valuable pedagogical bridge between "the software gives me a standard error" and "here is exactly where that standard error comes from."

1.19.2 Statistical model

As in Case Study 2, $Y_i \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with log-likelihood $\ell(\mu, \sigma)$ as above. The (observed) Fisher information for μ is the corresponding diagonal entry of the negative Hessian,

$$\mathcal{I}(\hat{\mu}) = - \left. \frac{\partial^2 \ell(\mu, \sigma)}{\partial \mu^2} \right|_{(\hat{\mu}, \hat{\sigma})}, \quad \widehat{\text{SE}}(\hat{\mu}) = \sqrt{\mathcal{I}(\hat{\mu})^{-1}},$$

and the asymptotic 95% confidence interval is

$$\hat{\mu} \pm z_{0.975} \cdot \widehat{\text{SE}}(\hat{\mu}), \quad z_{0.975} = \Phi^{-1}(0.975) \approx 1.96.$$

1.19.3 Python implementation

```

1 import pandas as pd
2 import numpy as np
3 import math
4 from statsmodels.base.model import GenericLikelihoodModel
5 from scipy.stats import norm
6
```

```

7  # --- Load data ---
8  df    = pd.read_excel("M3 Salary Data.xlsx", sheet_name="
      DataTable - 1")
9  data = df["McGill"].to_numpy()          # n = 100 McGill
      starting salaries
10
11 # --- Custom likelihood model (same structure as Case Study 2)
    ---
12 class MLE(GenericLikelihoodModel):
13     def __init__(self, data):
14         self.y = data
15         super().__init__(endog=np.zeros(1))
16         self.nparams = 2
17         self._set_extra_params_names(["mean", "std"])
18         self._eps = 1e-9
19
20     def loglike(self, params):
21         mean, std = params
22         if not (self._eps < std):
23             return -np.inf
24         return norm.logpdf(self.y, loc=mean, scale=std).sum()
25
26 # --- Fit ---
27 model = MLE(data)
28 result = model.fit(start_params=[0, 10], method="bfgs", maxiter
      =200)
29 print(result.summary())
30
31 # --- Extract point estimate ---
32 MLE_mean = result.params[0]      # mu_hat
33
34 # --- Observed Fisher information for mu (negative Hessian,
      entry [0,0]) ---
35 I_mean = -result.model.hessian(result.params)[0, 0]
36 SE_mean = math.sqrt(1 / I_mean)  # SE(mu_hat) = sqrt(I^{-1})
37
38 # --- 95% confidence interval via asymptotic normality ---
39 CV_LCL = norm.ppf(0.025)        # -1.96
40 CV_UCL = norm.ppf(0.975)        # +1.96
41
42 Interval_LCL = MLE_mean + CV_LCL * SE_mean
43 Interval_UCL = MLE_mean + CV_UCL * SE_mean
44
45 print(Interval_LCL)
46 print(Interval_UCL)

```

Implementation notes

- `result.model.hessian(result.params)` returns the 2×2 Hessian of $\ell(\mu, \sigma)$ evaluated at the MLE $(\hat{\mu}, \hat{\sigma})$; index `[0,0]` isolates $\partial^2 \ell / \partial \mu^2$.

- The Hessian of a log-likelihood is negative (semi-)definite at a maximum, so `-hessian[0,0]` is positive and can be inverted and square-rooted safely.
- `norm.ppf(0.025)` and `norm.ppf(0.975)` return the standard Normal critical values ∓ 1.96 used to build a two-sided 95% interval.
- This is exactly the calculation `statsmodels` performs internally to populate the `std err` and `[0.025, 0.975]` columns of `result.summary()` — reproducing it manually demystifies the summary table for students.

1.19.4 Output

```

1 Optimization terminated successfully.
2 Current function value: 1225.757504
3
4          coef      std err          z      P>|z|      [0.025      0.975]
5 -----
6 mean    1.09e+05    5095.818     21.396     0.000     9.9e+04    1.19e+05
7 std      5.096e+04   3603.958     14.140     0.000     4.39e+04     5.8e+04
8
9 I_mean    = 3.8510e-08
10 SE_mean   = 5095.818221111117
11 Interval_LCL = 99040.68285648646
12 Interval_UCL = 119015.92322676796

```

1.19.5 Interpretation

The point estimate $\hat{\mu} = \$109,028.30$ matches the sample mean, and the manually computed standard error, $SE(\hat{\mu}) = 5,095.82$, matches `result.summary()` exactly. The resulting 95% confidence interval,

$$CI_{95\%}(\mu) = (\$99,040.68, \$119,015.92),$$

also matches the `[0.025, 0.975]` columns of the summary table to the reported precision. This confirms, step by step, that the "black box" standard errors reported by `statsmodels` are nothing more than $1/\sqrt{\mathcal{I}(\hat{\theta})}$, i.e. the inverse square root of the observed Fisher information — a satisfying way to close the loop between the abstract asymptotic theory in Section 1 and the numbers printed by the software.

1.20 Case Study 4: A Pooled (Joint) Likelihood Across Two Independent Samples

1.20.1 Setting and data

We now use *both* the McGill and Toronto salary columns ($n_1 = n_2 = 100$ each) from the same workbook. Suppose a researcher hypothesizes that, despite coming from two different programs, starting salaries at both schools are generated by a *common* Normal distribution $\mathcal{N}(\mu, \sigma^2)$ — i.e., the two samples are pooled to estimate a single pair of parameters. This is a natural stepping stone toward likelihood-ratio tests that compare a "common parameters" model against a "separate parameters per group" model.

1.20.2 Statistical model

Let $\{y_{1i}\}_{i=1}^{n_1}$ (McGill) and $\{y_{2j}\}_{j=1}^{n_2}$ (Toronto) be two *independent* samples, both assumed $\sim \mathcal{N}(\mu, \sigma^2)$ under the shared parameterization. Because the two samples are independent, the joint likelihood factorizes and the joint log-likelihood is simply the *sum* of the two individual log-likelihoods:

$$\ell(\mu, \sigma) = \underbrace{\sum_{i=1}^{n_1} \log f(y_{1i} \mid \mu, \sigma)}_{\ell_1(\mu, \sigma)} + \underbrace{\sum_{j=1}^{n_2} \log f(y_{2j} \mid \mu, \sigma)}_{\ell_2(\mu, \sigma)}.$$

Maximizing $\ell(\mu, \sigma) = \ell_1 + \ell_2$ jointly over the *same* (μ, σ) yields estimates that combine information from both samples — effectively treating the pooled $n_1 + n_2 = 200$ observations as one sample, but implemented explicitly as a sum of two log-likelihood contributions, which generalizes directly to more complex multi-group or panel-data likelihoods.

1.20.3 Python implementation

```

1  import pandas as pd
2  import numpy as np
3  from statsmodels.base.model import GenericLikelihoodModel
4  from scipy.stats import norm
5
6  # --- Load data ---
7  df = pd.read_excel("M3 Salary Data.xlsx", sheet_name="
      DataTable - 1")
8  data1 = df["McGill"].to_numpy()           # sample 1, n1 = 100
9  data2 = df["Toronto"].to_numpy()         # sample 2, n2 = 100
10
11 # --- Custom model: joint likelihood of two independent samples
    ---
12 class MLE(GenericLikelihoodModel):
13     def __init__(self, data1, data2):
14         self.data1 = data1
15         self.data2 = data2
16         super().__init__(endog=np.zeros(1))
17         self.nparams = 2
18         self._set_extra_params_names(["mean", "std"])
19         self._eps = 1e-9
20
21     def loglike(self, params):
22         mean, std = params
23         if not (self._eps < std):
24             return -np.inf
25         l11 = norm.logpdf(self.data1, loc=mean, scale=std).sum
            ()
26         l12 = norm.logpdf(self.data2, loc=mean, scale=std).sum
            ()
27         return l11 + l12      # sum of independent log-likelihoods
28
29 # --- Fit jointly over the pooled parameters ---

```

```

30 model = MLE(data1, data2)
31 result = model.fit(start_params=[0, 10], method="bfgs", maxiter
    =200)
32 print(result.summary())

```

Implementation notes

- The constructor now stores *two* data arrays, `data1` and `data2`, but the model still has only two free parameters, (`mean`, `std`), shared across both samples.
- `loglike` returns `l11 + l12`: summing independent log-likelihoods is the general recipe for building a joint likelihood across any number of independent data sources (multiple groups, panels, or even different distributional families for different sources).
- This design pattern extends directly to a *likelihood-ratio test*: fit the pooled model here (2 parameters), fit two separate per-group models (2 parameters each, 4 total), and compare $2 \times (\ell_{\text{unrestricted}} - \ell_{\text{restricted}})$ against a χ^2 reference distribution.

1.20.4 Output

```

1 Optimization terminated successfully.
2 Current function value: 2448.229068
3 Iterations: 128
4 Function evaluations: 155
5 Gradient evaluations: 152
6
7      coef      std err      z      P>|z|      [0.025      0.975]
8  -----
9 mean    1.157e+05    3545.387    32.641    0.00    1.09e+05    1.23e+05
10 std     5.014e+04    2508.301    19.989    0.00    4.52e+04    5.51e+04

```

1.20.5 Interpretation

The pooled estimate $\hat{\mu} = \$115,700$ lies between the McGill-only mean (\$109,029, Case Study 3) and the (unreported here, but higher) Toronto-only mean, reflecting the fact that the model forces a single location parameter onto both programs. Notice the standard error on the pooled mean (3,545) is smaller than the McGill-only standard error (5,096 from Case Study 3): pooling $n_1 + n_2 = 200$ observations under the (possibly wrong) assumption of a common mean mechanically shrinks the standard error by roughly a factor of $\sqrt{2}$, illustrating both the efficiency gain from pooling *and* the risk of masking real between-group differences if the common-parameter assumption does not hold in practice.

1.21 Summary Table

1.22 Discussion Questions for Students

1. In Case Study 1, is $\hat{p} = 0.60$ statistically distinguishable from a fair coin ($p = 0.5$)? Construct the relevant Wald test using $\hat{p}$ and its standard error.

	Data	Model	n	Key MLE result
1	Coin flips	Bernoulli(p)	100	$\hat{p} = 0.600$ (SE = 0.049)
2	Auto MPG	$\mathcal{N}(\mu, \sigma^2)$	392	$\hat{\mu} = 23.446$, $\hat{\sigma} = 7.795$
3	McGill salary	$\mathcal{N}(\mu, \sigma^2) + 95\%$ CI	100	$\hat{\mu} = \$109,028$, CI = (\$99,041, \$119,016)
4	McGill & Toronto (pooled)	Joint $\mathcal{N}(\mu, \sigma^2)$	200	$\hat{\mu} = \$115,700$, $\hat{\sigma} = \$50,140$

Table 1.14: Overview of the four MLE case studies, all fitted with `GenericLikelihoodModel` and the BFGS optimizer.

2. In Case Study 2, why does the MLE of σ use an n -denominator rather than $n - 1$? What does this imply about the bias of $\hat{\sigma}_{\text{MLE}}^2$ in small samples?
3. In Case Study 3, reproduce the confidence interval for σ (rather than μ) using the (1, 1) entry of the Hessian, and compare it to the [0.025, 0.975] row for `std` in `result.summary()`.
4. In Case Study 4, fit *separate* (μ, σ) models to McGill and Toronto individually, and use a likelihood-ratio test to formally test whether pooling the two samples under a common mean is statistically justified.

1.23 Closing synthesis

Maximum likelihood provides one coherent workflow across binary and continuous data: specify the sampling model, add the observation-level log-likelihood contributions, maximize over admissible parameters, and quantify local uncertainty using information. The estimate is inseparable from its assumptions. A precise optimizer cannot repair incorrect coding, non-independence, an implausible distribution, or an unjustified pooling restriction.

Research habit. For every fitted model, report the data-generating assumption, the parameterization, the estimate, its standard error or confidence interval, convergence evidence, and at least one diagnostic that challenges the model rather than merely confirming it.

References
